

A top-down view of a workspace. In the upper left, a portion of a silver laptop is visible, showing the keyboard and trackpad. Below it, a tablet displays the text 'BAIT3153 Software and Project Management (SPM)'. To the right of the tablet is a black cup of coffee on a matching saucer. The background is a dark, textured desk surface.

Chapter 1

BAIT3153 Software and
Project Management (SPM)

Project Management Concepts

Chapter 1 Table of Contents

1.1 Introduction to Project Management

1.2 Project Life Cycle

1.3 Project Appraisal

**1.4 Knowledge and Attributes of a Project
Manager**

1.5 Project Management Failure



1.1 An Introduction to Project Management

- What is Project Management?
- What is a Project?
- Characteristics of Projects
- Software vs Other Types of Projects
- Constraints in Projects

What is Project Management?

1.1 Introduction to Project Management

A discipline that involves:

- **Planning** - deciding what to be done (Project Initiation Document [PID] needs to be created)
- **Organising** - making arrangement
- **Staffing** - select personnel
- **Directing** - give instructions
- **Monitoring** - check on progress
- **Controlling** - remedy hold-ups
- **Innovating** - coming up with new solutions

Planning

Organising

Staffing

Directing

Controlling

Innovating

What is Project Management?

1.1 Introduction to Project Management

A discipline that involves:

- **Planning** - deciding what needs to be done,
(then...)
- **Organising** - make arrangement
When you want to do it? How you want to do it? Who
- **Staffing** - select personnel
- **Directing** - give instructions
- **Monitoring** - monitor progress
- **Controlling** - remedy hold-ups
- **Innovating** - coming up with new solutions

Planning

Organising

Staffing

Directing

Controlling

Innovating

What is a Project?

1.1 Introduction to Project Management

- “A **specific** plan or design”
- “A **planned** undertaking”
- “A **large** undertaking”

Examples of project

- The ECRL (East Coast Rail Link) —
- Developing the Huawei's Harmony OS
- _____



Characteristics of Projects

1.1 Introduction to Project Management

- Non-routine (e.g. FYP)
- Planning is required (e.g. what needs to be done etc...)
- Specific objectives to be met
(e.g. Develop a student attendance system to reduce the amount of paper consumption by 99%)
- Pre-determined time span (e.g. FYP 9 months)
- Work is carried out for someone, i.e. clients
- Involves several specialisms (e.g. database specialists, UI specialist, Network specialist, etc)
- Involves several phases (e.g. analysis, design, code, test, implement, _____)
- Resources constraints is common
(e.g. Time budget staffs)



Software vs Other Types of Projects

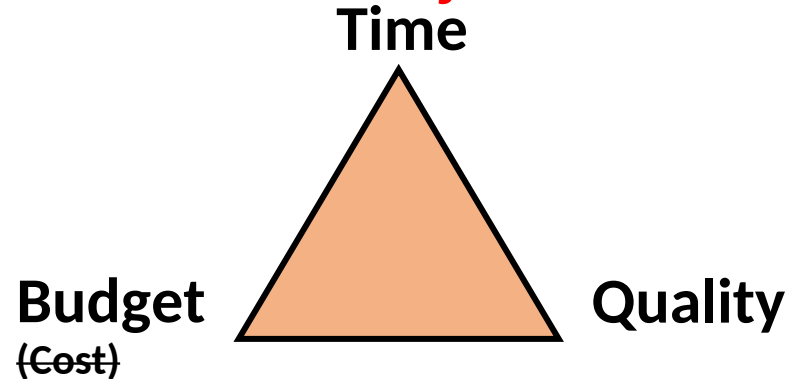
1.1 Introduction to Project Management

	Software Project (e.g. developing XYZ operating system)	Other Types of Project (e.g. construction project)
Invisibility	progress <u>is not</u> immediately visible (<i>no physical form</i>)	progress <u>is</u> immediately visible (<i>have physical form</i>)
Conformity	have to conform to <u>clients</u> requirements	have to conform to <u>physical laws</u> (e.g. gravity law)
Complexity	Have <u>more complexity</u> than other engineered artefacts	<u>less complex</u> than software project
Flexibility	<u>Easier</u> to change	<u>Difficult</u> to change once it is completed

Constraints in Projects

1.1 Introduction to Project Management

- Project managers need to face 3 major constraints:



- Project managers' mantra:
To develop a system **[on time, within budget and to the required quality]**

FYP students, your mantra = To complete FYP project scope on time and to the required quality

Challenges in Software Projects

Challenges faced by **project manager**

- Poor estimates and plans
- Lack of quality standards and measures
- Lack of guidance about making organisational decisions
- Lack of techniques to make progress visible
- Poor role definition – who does what?
- Incorrect success criteria
- **Impossible deadlines**
- **Scope changes (affect ?)**
- **Resource constraints**

Challenges faced by **developers**

- Inadequate specification of work
- Management ignorance of IT
- Lack of knowledge of application area
- Lack of standards
- Lack of up-to-date documentations
- **Preceding activities not completed on time (This will cause what problem?)**
- Narrow scope of technical expertise
- Changing user requirement
- **Deadline pressure**
- **Lack of skill**
- Lack of communication

Challenges faced by **customers**

- **The software project is behind schedule (problem = ?)** (e.g. need to wait for 1 more month then only can use the software)
- **The developed software no longer meets the needs of the organization**
- **The cost of the project has exceeded its allotted budget**



1.2 Project Life Cycle

- Generic Project Process Model
- Project Life Cycle Processes
- RFP and Proposals

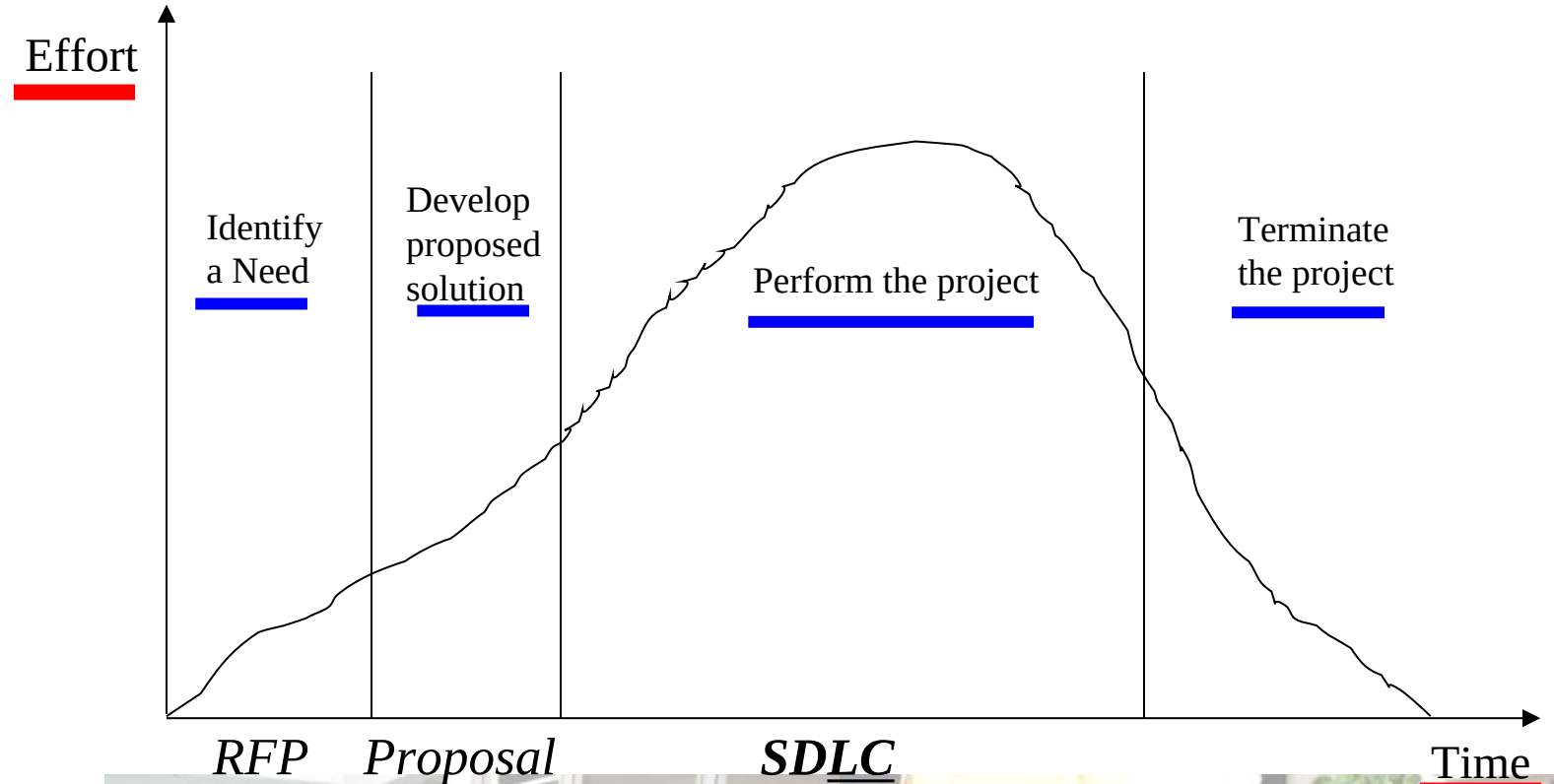
Generic Project Process Model

1.2 Project Life Cycle

The project life cycle

4 stages:

- Identify a need
- Develop proposed solution
- Perform the project
- Terminate the project



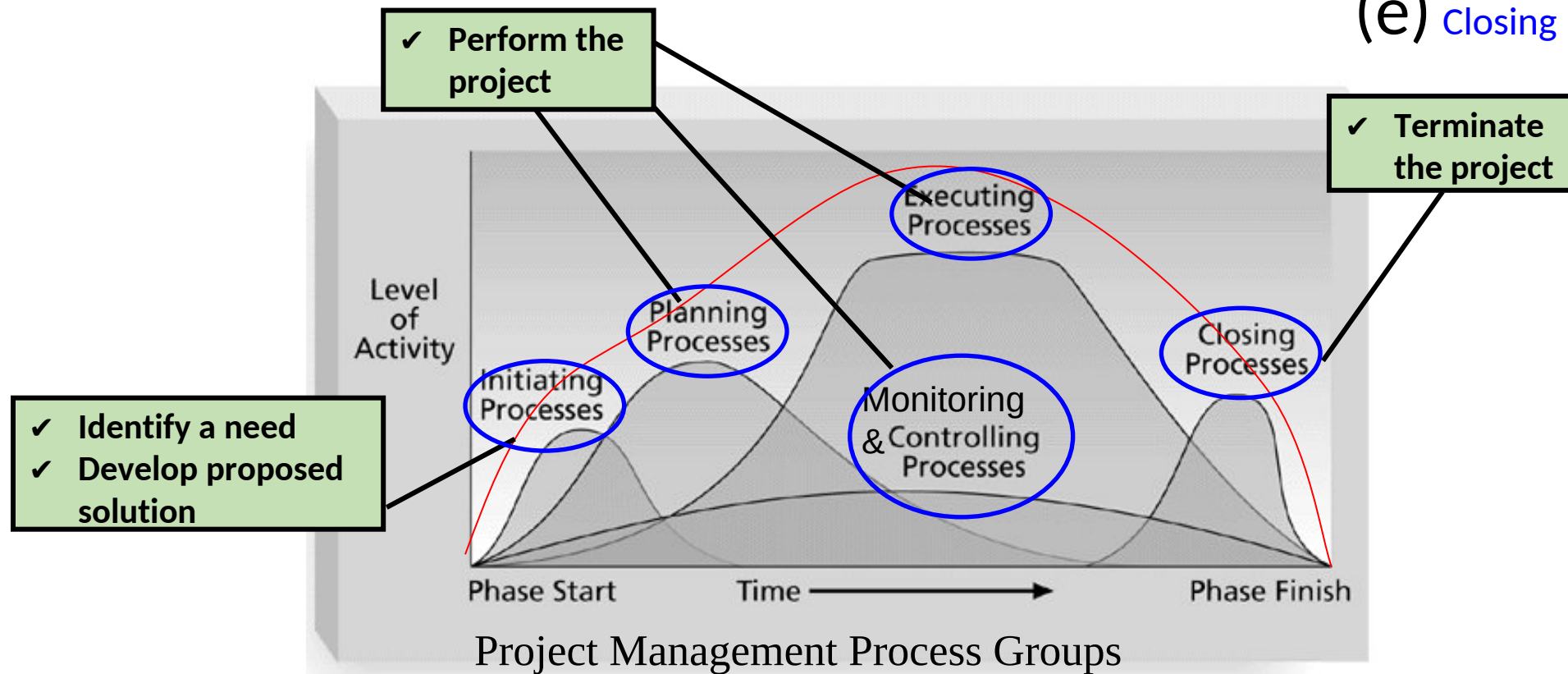
*A good project is one that has successfully fulfilled the identified **need**!*

⌘ Ana - des - cod - tes - imp ✉

Project Life Cycle Processes

1.2 Project Life Cycle

- (a) Initiating Processes
- (b) Planning Processes
- (c) Executing Processes
- (d) Monitoring & Controlling Processes
- (e) Closing Processes



(a) Initiating

Project Life Cycle Processes

- **Conduct a kick-off meeting** to establish customer's needs, goals, requirements and expected benefits.
- **Then 1 or 2 more meetings** to streamline the project requirements
- **Output: Project Charter**

*Note: a project charter **is a formal project** authorization which **documents** the shared understanding of a **project's scope**, development and **objectives**, while also defining the **roles and responsibilities of each party involved**. (<https://thedigitalprojectmanager.com/project-charter/>)*



(b) Planning

Project Life Cycle Processes

- **Create a Project Management Plan (PMP)** and kept it up-to-date as the project progresses.
- PMP contains sufficient details to monitor and exercise control over the project execution e.g. **project schedules, milestones, project team members, and all other project related information.**
- In maintenance projects, the detailed schedule is dynamically updated as and when maintenance requests are received and delivered.



(c) Executing

Project Life Cycle Processes

- Once the plan is in place - **execute the project**
- The Project Manager evaluates adherence to the project planning process and the work products and services of the process to the applicable requirements, objectives, and standards, and addresses noncompliance.
- By ensuring adherence to the plan, information will be available to accurately identify issues and problems in their early stages and minimize risks.
- **Output: work results**
(Give some examples of FYP work results = ? _____)



(d) Monitoring & Controlling

Project Life Cycle Processes

- During the project execution, the Project Manager monitor the progress against the plan (**means** check if “project progress” = “What has been planned”
e.g. tasks completed on time?, if ans = “No”...)
- For this, the Project Manager gathers the information needed, analyzes it and identifies deviations that are significant.
- **Take corrective actions** to solve identified issues. This may include re-planning and establishing new agreements. (e.g. healthy if project is on time and within budget)
(What is the overall health of your FYP?)
- **Output:** overall health of project



(e) Closing

Project Life Cycle Processes

- **When the project is completed** the Project Manager gathers the project related data and documents.
- **Handover the project's product to the customer**
(e.g. completed software, _____)
- **Conduct post project review (self reflection).**
- **Redeploy project resources**

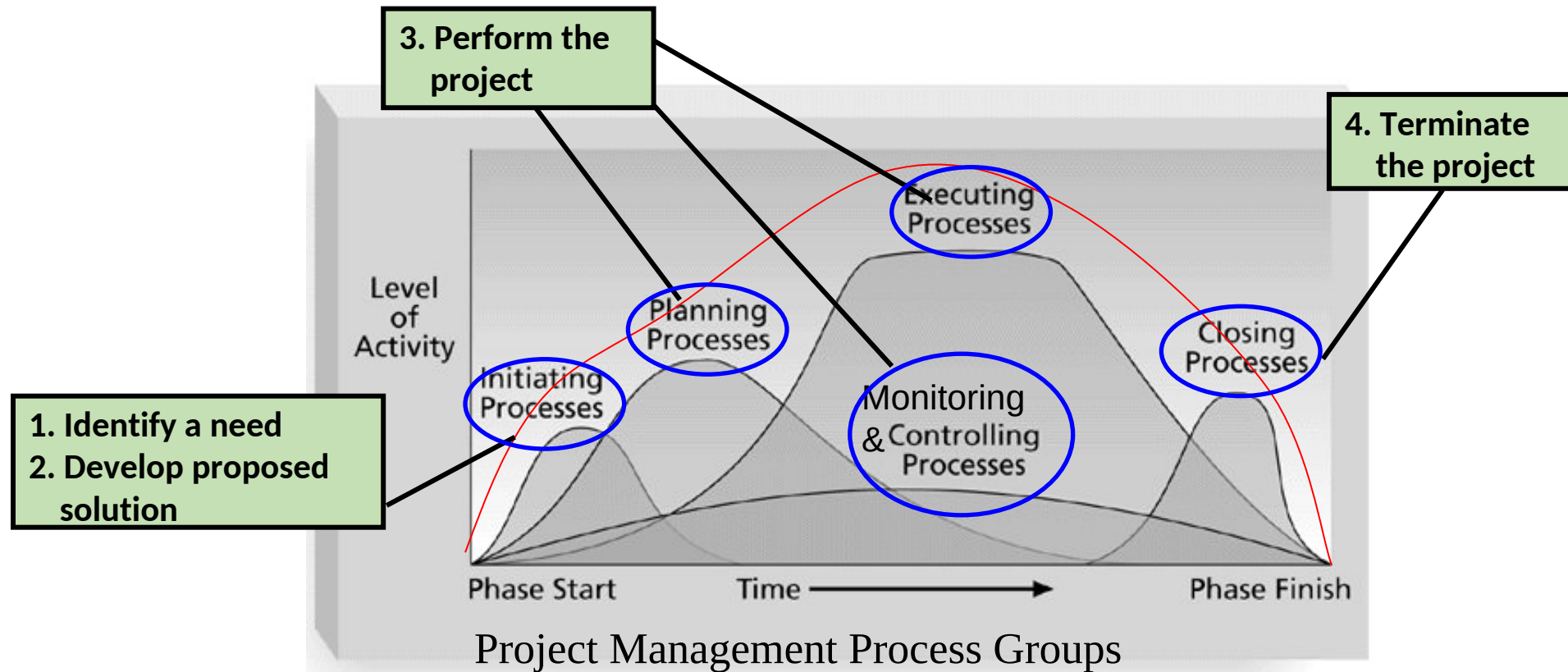


Quick Review

Project Life Cycle Processes

1.2 Project Life Cycle

- (a) Initiating Processes
- (b) Planning Processes
- (c) Executing Processes
- (d) Monitoring & Controlling Processes
- (e) Closing Processes



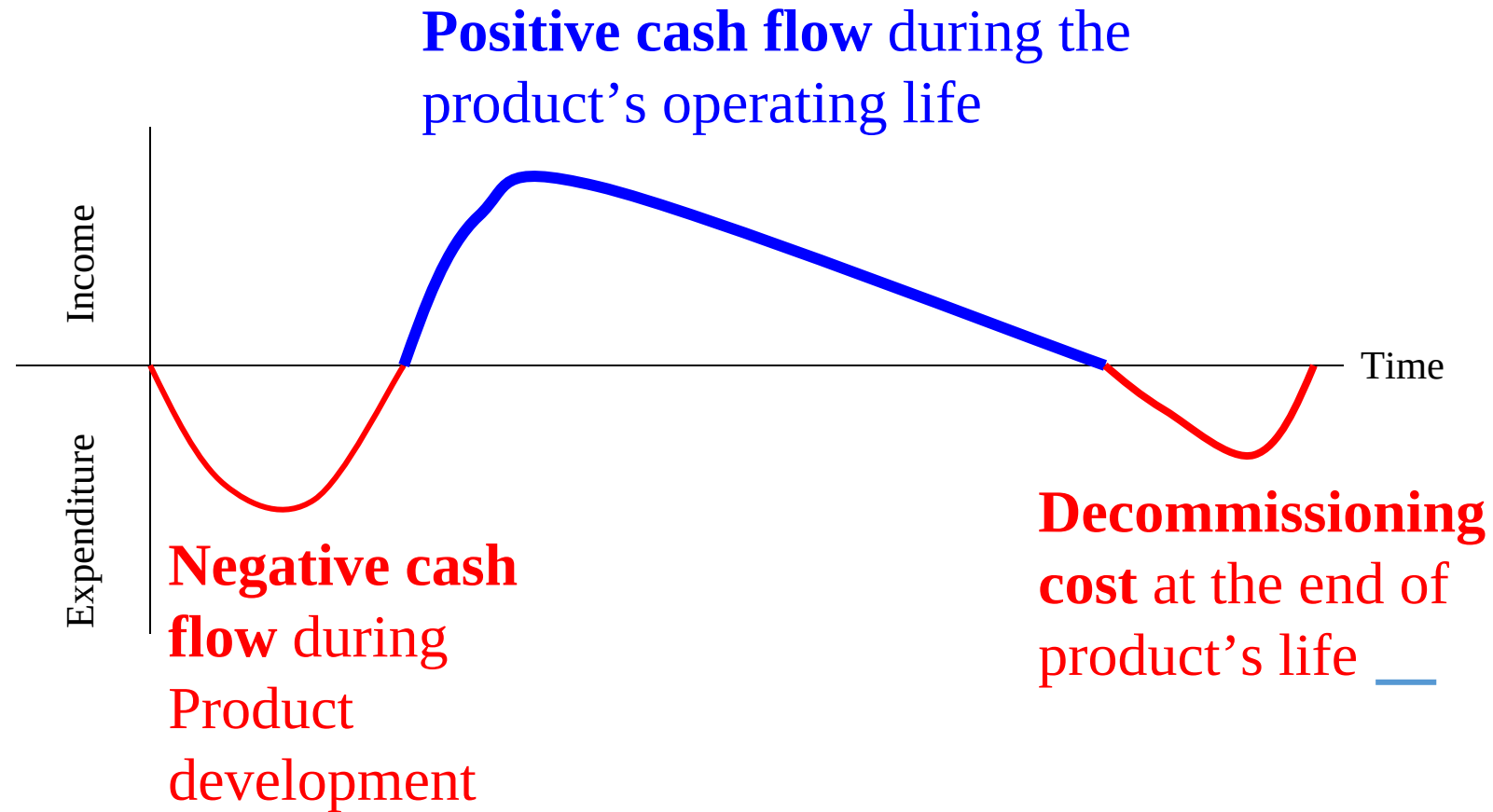


1.3 Project Appraisal

- Cash Flow Forecasting
- Cost & Benefit Analysis
- Financial Techniques
 - Net Profit
 - Payback Period
 - Return on Investment (ROI)
 - Net Present Value (NPV)
- Weighted Scoring Model

Things to do **before** project starts to **save** the organisation from a bad decision or investment.

Cash Flow Forecasting



Cash Flow Forecasting is ...

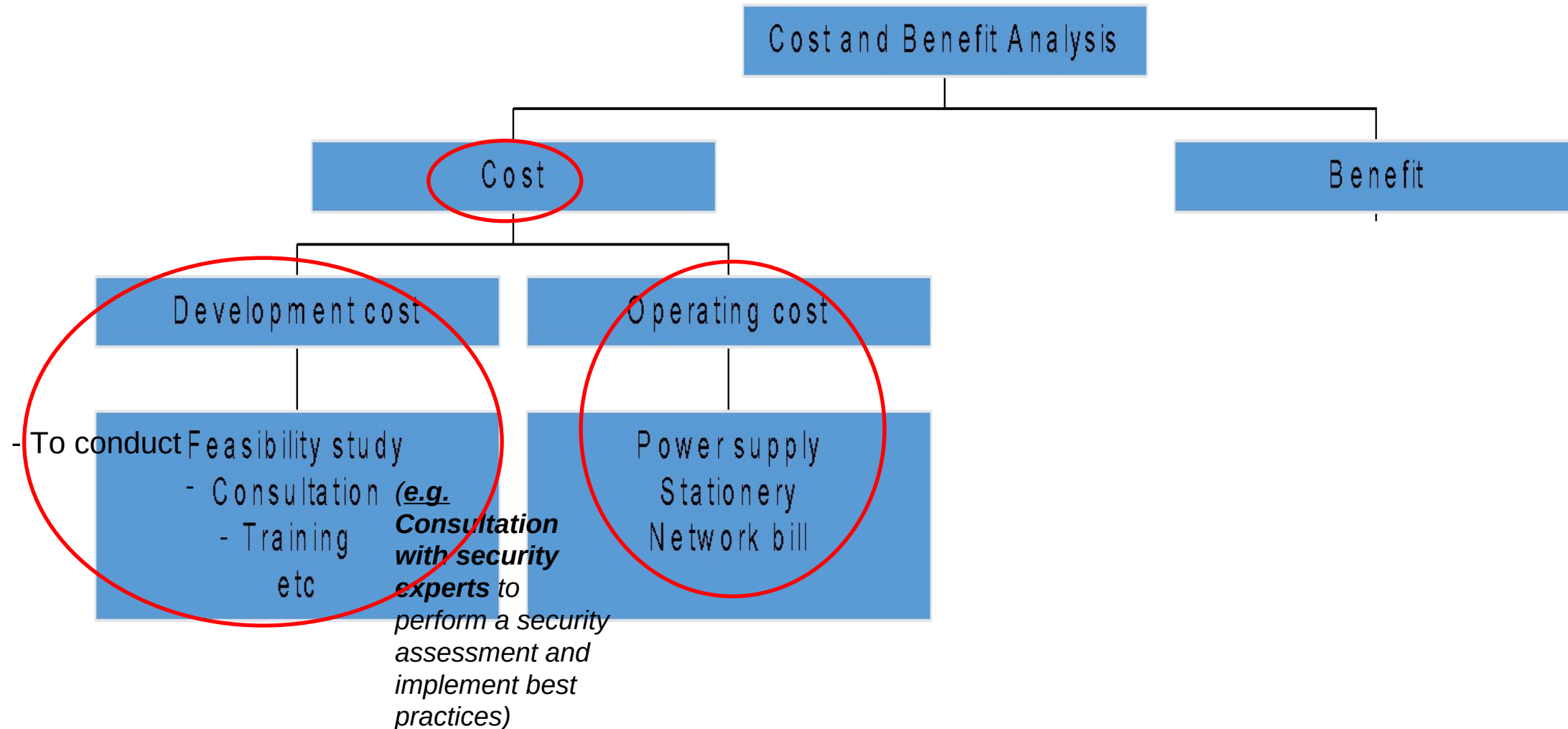


Cost & Benefit Analysis

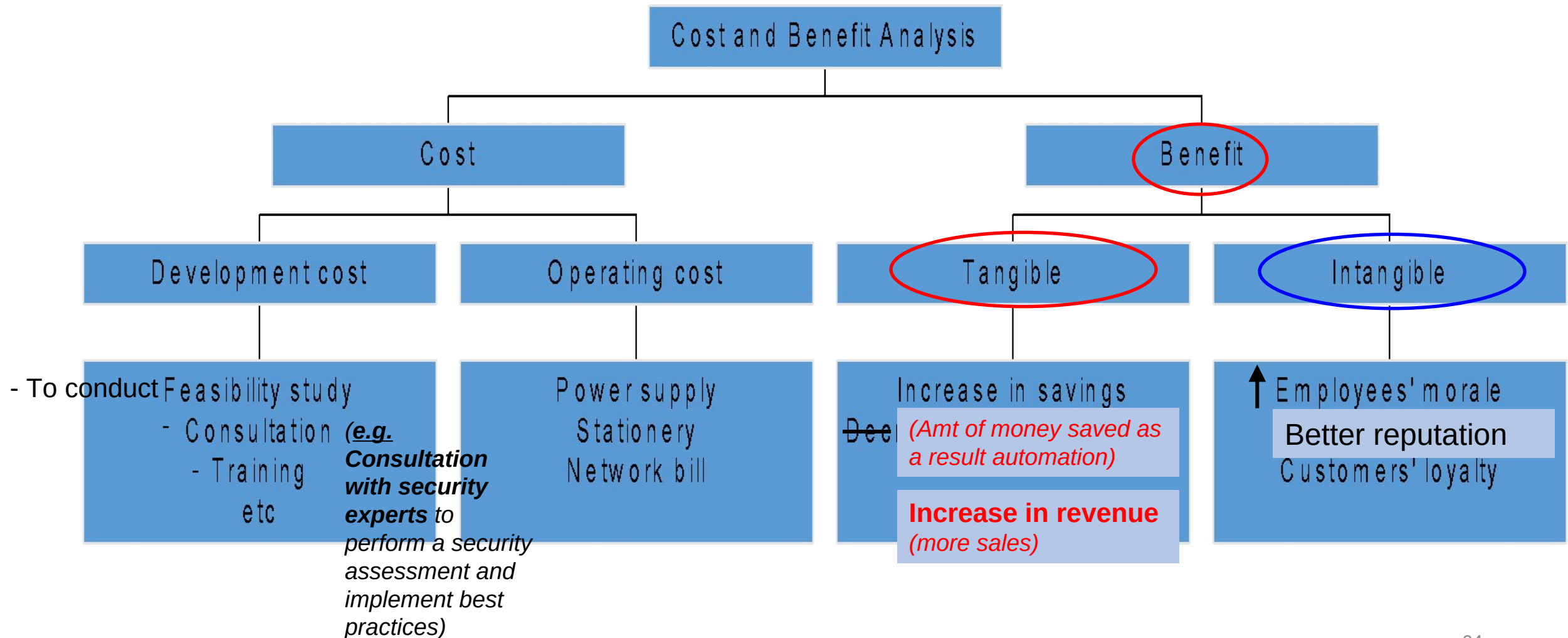
- A comparative assessment of all the benefits you anticipate from the project and all the costs required to introduce the project, perform it and support the changes resulting from it.
- Helps you to
 - Decide whether to undertake a project or decide which of several projects to undertake.
 - Frame appropriate project objectives.
 - Develop appropriate before and after measures of project success.
 - Prepare estimates of the resources required to perform the project work.



Cost & Benefit Analysis - Examples



Cost & Benefit Analysis - Examples





1.3 Project Appraisal

- Cash Flow Forecasting
- Cost & Benefit Analysis
- **Financial Techniques**
 - Net Profit
 - Payback Period
 - Return on Investment (ROI)
 - Net Present Value (NPV)

*Measure
project
profitability*



Net Profit (NP)

1.3 Project Appraisal

Net profit = total income - total costs (over the life of the project)

E.g. (negative sign = cost)



Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net Profit	50,000	100,000	50,000	75,000

Table 1.1 4 project cash flow projections – figures are end of year totals (RM)

Based on NP technique alone which project you would approve for development?



Net Profit

1.3 Project Appraisal

Year	<i>Project 1</i>	<i>Project 2</i>	<i>Project 3</i>	<i>Project 4</i>
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net Profit	50,000	100,000	50,000	75,000



Advantages

Easy to calculate and intuitive.

Disadvantages

- NP technique does not consider the time value of money (See Project 2, Year 5)
(*note: the value of a dollar received in the future is not the same as a dollar received today*)
- **Project 1 & 3** have equal cost & net profit but are both projects equally preferable?
To ans this question you need to analyse further
(**Project 1** has bulk of income occurs late in the life of the project. Having to wait longer for a return is a disadvantage. **Project 3** returns a steady income throughout its life. Thus, project 1 & 3 are not equally preferable)



Payback Period

1.3 Project Appraisal

- **Calculate the time needed to break even** or pay back the initial investment.
 - **Normally project with the shortest payback period will be chosen** on the basis that an organization will wish to minimize the time that a project is 'in debt'.
 - **How to calculate payback period?**
(Subtract each individual annual cash **inflow** from the initial cash **outflow**, until the payback period is achieved)
- 



Payback Period

1.3 Project Appraisal

What is the payback period for Project 1, 2, 3 and 4?



Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net Profit	50,000	100,000	50,000	75,000

Table 1.2 Four project cash flow projections – figures are end of year totals (RM)



Payback Period

1.3 Project Appraisal

What is the payback period for Project 1, 2, 3 and 4?

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net Profit	50,000	100,000	50,000	75,000



Table 1.2 Four project cash flow projections – figures are end of year totals (RM)

- Using payback period technique, **Project 3** has the shortest payback period. *(Project 3 breakeven in **early Y4**. Project 4 breakeven at the **end of Y4**)*
- But **Net Profit** analysis shows that **Project 2 and Project 4** are more profitable than Project 3.

Payback Period

1.3 Project Appraisal

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net Profit	50,000	100,000	50,000	75,000

IF project 3 is chosen, company will miss 75K in Project 4

Advantages:

- **Easy** to calculate and understand
- Projects with shorter payback periods are **generally less risky** (as they recover their investment faster, reducing the risk of insolvency)

Disadvantages:

- It **ignores** any profits generated after the payback period (e.g. compare Project 3 and project 4)
- **PP technique also does not consider the time value of money** (It does not discount future cash flows)



Return on Investment (ROI)

1.3 Project Appraisal

- Aka Accounting Rate of Return (ARR)
- Provides a way of comparing the net profitability to the investment required.



- **Formula:**
$$\text{ROI} = \frac{\text{average annual profit}}{\text{total investment}} \times 100$$

Return on Investment (ROI)

1.3 Project Appraisal

What is the ROI for Project 1, 2, 3 and 4?

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net Profit	50,000	100,000	50,000	75,000
ROI	?	?	?	?

Table 1.3 Four project cash flow projections – figures are end of year totals (RM)

Calculate ROI:

Project 1

$$\text{ROI} = \frac{\text{Avg annual profit}}{\text{Tot investment}} \times 100 = 10\%$$

Project 2

$$\text{ROI} = \frac{\text{Avg annual profit}}{\text{Tot investment}} \times 100 = 2\%$$

A vertical image on the left side of the slide shows a black and gold calculator, a pen, and several colorful paper clips (orange, yellow, pink) resting on a blue grid-patterned surface.

Return on Investment (ROI)

1.3 Project Appraisal

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net Profit	50,000	100,000	50,000	75,000
ROI	10%	2%	10%	12.5%

Table 1.3 Four project cash flow projections – figures are end of year totals (RM)

Which project is most worthwhile based on ROI alone?

A close-up photograph of a black and gold calculator with a green display screen, resting on a white surface. In the foreground, several colorful paper clips (orange, yellow, pink) are visible, along with a portion of a blue and white striped object, possibly a folder or a piece of paper.

Return on Investment (ROI)

1.3 Project Appraisal

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net Profit	50,000	100,000	50,000	75,000
ROI	10%	2%	10%	12.5%

Advantages:

- Simple and easy to calculate
- Return rate provides a clear indicator of the potential returns

Disadvantages:

- Does not consider the time value of money.
- High ROI but a long payback period may not be good.



Net Present Value (NPV)

1.3 Project Appraisal

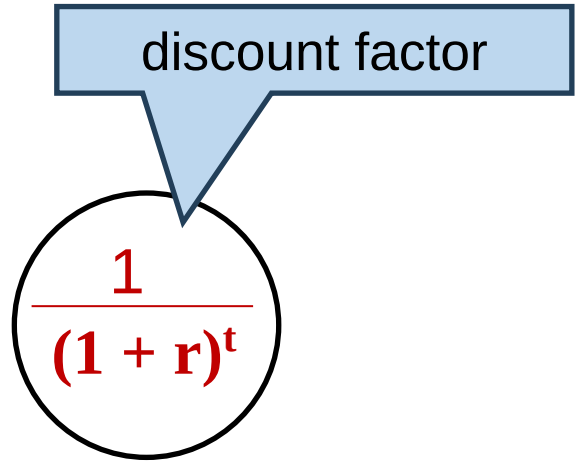
- NPV takes into account the time value of money.
 - Is a financial method used to determine the current/present value of all future cash flows generated by a project including the initial capital investment.
 - 2 views:
 - Receiving RM100 today is better than having to wait until next year to receive it.
 - The present value of RM100 in a year's time is RM91.
 - mean $Y1 \text{ (RM100)} = Y0 \text{ (RM91)}$, $Y0 = \text{Now}$, $Y1 = 1 \text{ year later}$ (assuming Discount rate = 10%)
 - Invest RM100 in a bank today and have RM110 in a year's time
- 

Net Present Value (NPV)

1.3 Project Appraisal

How to calculate the Present Value of a future cash flow ?

Present Value
of a future cash flow = Cash Flow in Year t X



A blue rectangular box labeled "discount factor" has a blue arrow pointing to a circle containing the formula $\frac{1}{(1 + r)^t}$.

$$\frac{1}{(1 + r)^t}$$

where

- r is the **discount rate** (expressed as a decimal value)
- t is the **number of years into the future** that the cash flow occurs

Net Present Value (NPV)

1.3 Project Appraisal

Example: Calculate the NPV for project 1
Assuming **discount rate = 10%**

Year	Project 1 Cash Flow (RM)	Discount Factor	Discounted cash flow or Present Value
0 (NOW)	-100,000	1.0000	-100,000
1	10,000	0.9091	9,091
2	10,000	0.8264	8,264
3	10,000	0.7513	7,513
4	20,000	0.6830	13,660
5	100,000	0.6209	62,090
			NPV: RM618

$$1/(1+0.10)^0$$

Present value of future cash flow

Future cash flow

$$1/(1+0.10)^5$$

A black and gold calculator is visible on the left side of the slide, with several colorful paper clips (orange, yellow, pink) resting on a blue grid pattern at the bottom left.

Net Present Value (NPV)

1.3 Project Appraisal

Exercise

Using a **10% discount rate**, calculate the NPV for **Project 3**.

Year	Project 3 Cash Flow (RM)	Discount Factor @ 10%	Discounted cash flow or Present Value
0	-100,000	1.0000	-100,000
1	30,000	0.9091	27,273
2	30,000	0.8264	24,792
3	30,000	0.7513	22,539
4	30,000	0.6830	20,490
5	30,000	0.6209	18,627
Net Profit	50,000		NPV: RM13,721

Based on NPV alone, is **Project 3** more beneficial to pursue than **Project 1** (NPV = **RM618**)?

Yes,

i) A **positive NPV** indicates the project is **profitable** [PV of cash inflows (benefits) > PV of cash outflows (costs) over the 5-year period]

ii) **Project 3 NPV is > Project 1** (P3 is > profitable than P1)

(assuming all other factors are equal)



Net Present Value (NPV)

1.3 Project Appraisal

- The **discount rate** can be thought of as a **target rate of return**.
 - If we set a target rate of return of 15% we would reject any project that did not display a **$NPV > 0$**
- 



Net Present Value (NPV)

1.3 Project Appraisal



Year	Project 2 Cash Flow (RM)	Discount Factor @ 10%	Discounted Cash Flow
0	-1,000,000	1.0000	-1,000,000
1	200,000	0.9091	181,820
2	200,000	0.8264	165,280
3	200,000	0.7513	150,260
4	200,000	0.6830	136,600
5	300,000	0.6209	186,270
Net Profit	100,000		NPV: -179,770

Is it worth to pursue **Project 2**?

No : **Project 2's** NPV is negative which indicates that it does not achieve the 10% target rate of return.

Project 2's **PV of cash outflows(costs) > PV of cash inflows (benefits)** over the 5-year period]. It is a net loss.

Exercise: Using a discount rate of 9%, calculate the NPV and ROI for **Project B**.

Project B	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	60,000	60,000	50,000	
Discount Factor					
Discounted Costs					
Benefits	0	80,000	90,000	180,000	
Discount Factor					
Discounted Benefits					
Discounted benefits – costs	(100,000)				
Cumulative benefits - costs	(100,000)				
ROI					

Exercise: Using a discount rate of 9%, calculate the NPV and ROI for Project B.

$$\text{ROI} = \text{NPV} / \text{Discounted Cost} \times 100\%$$

Project B	Year 0	Year 1	Year 2	Year 3	Total
Costs	100,000	60,000	60,000	50,000	
Discount Factor	1.0000	0.9174	0.8417	0.7722	
Discounted Costs	100,000	55,044	50,502	38,610	244,156
Benefits	0	80,000	90,000	180,000	
Discount Factor	1.0000	0.9174	0.8417	0.7722	
Discounted Benefits	0	73,392	75,753	138,996	288,141
Discounted benefits – costs	(100,000)	18,348	25,251	100,386	NPV = 43,985
Cumulative Discounted benefits – costs	(100,000)	(81,652)	(56,401)	43,985	

$$\text{ROI} = \frac{\text{Total Discounted Benefits} - \text{Total Discounted Cost}}{\text{Total Discounted Cost}} \times 100\%$$
$$= (43,985 / 244,156) \times 100\% = 18.02\%$$



Quick Review

1.3 Project Appraisal

- Cash Flow Forecasting
- Cost & Benefit Analysis
- Financial Techniques
 - Net Profit
 - Payback Period
 - Return on Investment (ROI)
 - Net Present Value (NPV)

*Measure
project
profitability*

Things to
do _____
project
starts
to _____ the
organisatio
n from a
bad
decision or
investment.

L2: Quick Review of PreVL

1.1 Introduction to Project Management

What is PM?, What is a project? Characteristics of project, Software projects Vs other types of projects, Challenges in software projects

1.2 Project Life Cycle

4 stages, PLC processes

1.3 Project Appraisal

Cash flow forecasting, Cost & benefits analysis

Financial techniques: NP, PP, ROI, NPV

Non-financial technique: _____

1.4 Knowledge and Attributes of a Project Manager

1.5 Project Management Failure

Weighted Scoring Model

1.3 Project Appraisal

- Specify the relevant **criteria for project selection**

(NPV, ROI can be the criterion)

- Weight** the importance of each criterion.

Assign a score
(0 to 100) to
each criterion

- **High score** means
the project strongly
meet the specified
criterion

- **Low score** means the
project is less likely to
meet the specified criterion

Criteria (for project selection)	Weight (%)	Project 1	Project 2	Project 3
Support business objectives	10%	80	90	50
Availability of resources	10%	70	70	50
Duration < 12 months	10%	70	90	50
Low risk	5%	50	50	50
High probability of project success	15%	90	90	50
Good ROI (above 40%)	50%	80	90	50
Weighted Project Scores	100%	78	86	50

50% means
Good ROI is a
very important
criteria

Weighted Project Scores for Project 2: $(10\% \times 90) + (10\% \times 70) + (10\% \times 90) + (5\% \times 50) + (15\% \times 90) + (50\% \times$

$90) = 86$

Choose project with the highest score!

1.4 Knowledge & Attributes of a Project Manager (PM)

- Knowledge of a PM
- Attributes of a PM



Knowledge of a PM

1.4 Knowledge & Attributes of a Project Manager

10 Knowledge Areas



	PLC/S1 + 2 Initiating	PLC/S3 (PERFORM THE PROJECT) Planning		Executing	M&C	PLC/S4 Closing
Integration	Develop Project Charter	Develop PM Plan		Direct & Manage Project Work	- M&C Project Work - Perform Integrated Change Control	Close Project or Phase
Scope		- Plan Scope Management - Collect Requirements - Define Scope - Create WBS			- Validate Scope - Control Scope	
Time		- Plan Schedule Management - Define Activities - Sequence Activities - Estimate Activity Resources - Estimate Activity Durations - Develop Schedule			Control Schedule	
Cost		- Plan Cost Management - Estimate Costs - Determine Budget			Control Costs	
Quality		Plan Quality Management		Perform Quality Assurance	Control Quality	
HR		Plan HR Management		- Acquire Project Team - Develop Project Team - Manage Project Team		
Communications		Plan Communications Management		Manage Communications	Control Communications	
Risk		- Plan Risk Management - Identify Risks - Perform Qualitative Risk Analysis - Perform Quantitative Risk Analysis - Plan Risk Responses			Control risks	
Procurement		Plan Procurement Management		Conduct Procurements	Control Procurements	Close Procurements
Stakeholder	Identify Stakeholders	Plan Stakeholder Management		Manage Stakeholder Expectation	Monitor Stakeholder Engagement	

Project Scope Management

1.4 Knowledge & Attributes of a Project Manager

- **Define and control what work is and is not included in a project.**
(Exclusion)
- ensures that the project team and stakeholders have the same understanding of deliverable (products) and what processes the project team will use to produce them.
- **6 main processes:**
 1. Plan scope management
 2. Collect requirements
 3. Define scope
 4. Create WBS
 5. Validate scope
 6. Control scope



Project Scope Management

1.4 Knowledge & Attributes of a Project Manager



1. Plan scope management

Scope defines what work will be done and what won't.

Scope mgmt define how scope will be managed throughout the project *to prevent uncontrolled changes*

Project Scope Management

1.4 Knowledge & Attributes of a Project Manager



2. Collect Requirements

Gather and document the needs of **stakeholders** (customer/project sponsor) and translate them into project requirements.

Project Scope Management

1.4 Knowledge & Attributes of a Project Manager



3. Define Scope

- Review the project charter, requirements documents, and organizational process assets to **create a scope statement (what are included and excluded)**
- **Adding more information to the scope statement as requirements are developed and change requests are approved.**

Project Scope Management

1.4 Knowledge & Attributes of a Project Manager



4. Create A Work Breakdown Structure (WBS)

- Subdivide the major project activities into smaller, more manageable components.
- **WBS is a hierarchy of activities.**
- Can be organized around project products, project phase, or using the project management process groups.
- **Guideline:** breakdown the work until **each task** can be performed by **1 person** within **8-80 hours (1-10 days)**

1.0 Concept

1.1 Evaluate current systems

1.2 Define requirements

1.2.1 Define user requirements

1.2.2 Define content requirements

1.2.3 Define system requirements

1.2.4 Define server owner requirements

1.3 Define specific functionality

1.4 Define risks and risk management approach

1.5 Develop project plan

1.6 Brief Web development team

2.0 Web site design

3.0 Web site development

4.0 Roll out

5.0 Support

Project Scope Management

1.4 Knowledge & Attributes of a Project Manager

Organized the **WBS** by
phases/stages in
OUTLINE
NUMBERING format

Q) There are how
many **phases or**
stages in this
WBS?

1.0 Concept

1.1 Evaluate current systems

1.2 Define requirements

1.2.1 Define user requirements

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Project Scope Management

1.4 Knowledge & Attributes of a Project Manager

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4.0 Roll out

5.0 Support

1. Arrange WBS in logical sequence
2. Indent sub-tasks

1.0 Concept

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1.2 Define requirements

1.2.1 Define user requirements

1.2.2 Define content requirements

1.2.3 Define system requirements

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Project Scope Management

1.4 Knowledge & Attributes of a Project Manager



5. Validate Scope

- To formalize acceptance of the project deliverables
- **Let Stakeholders** (customer/project sponsor) **inspect and formally accept deliverable** during this process.
- **If not acceptable, stakeholder usually request for changes**

1.0 Concept

1.1 Evaluate current systems

1.2 Define requirements

1.2.1 Define user requirements

1.2.2 Define content requirements

1.2.3 Define system requirements

1.2.4 Define server owner requirements

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4.0 Roll out

5.0 Support

Project Scope Management

1.4 Knowledge & Attributes of a Project Manager



6. Control Scope

Control changes to project scope throughout the life of the project.

- 1.0 Concept
 - 1.1 Evaluate current systems
 - 1.2 Define requirements
 - 1.2.1 Define user requirements
 - 1.2.2 Define content requirements
 - 1.2.3 Define system requirements
 - 1.2.4 Define server owner requirements
 - 1.3 Define specific functionality
 - 1.4 Define risks and risk management approach
 - 1.5 Develop project plan
 - 1.6 Brief Web development team
- 2.0 Web site design
- 3.0 Web site development
- 4.0 Roll out
- 5.0 Support

6.
Control
scope

Knowledge of a PM

1.4 Knowledge & Attributes of a Project Manager

10 Knowledge Areas



Project Time Management

1.4 Knowledge & Attributes of a Project Manager

7 main processes:

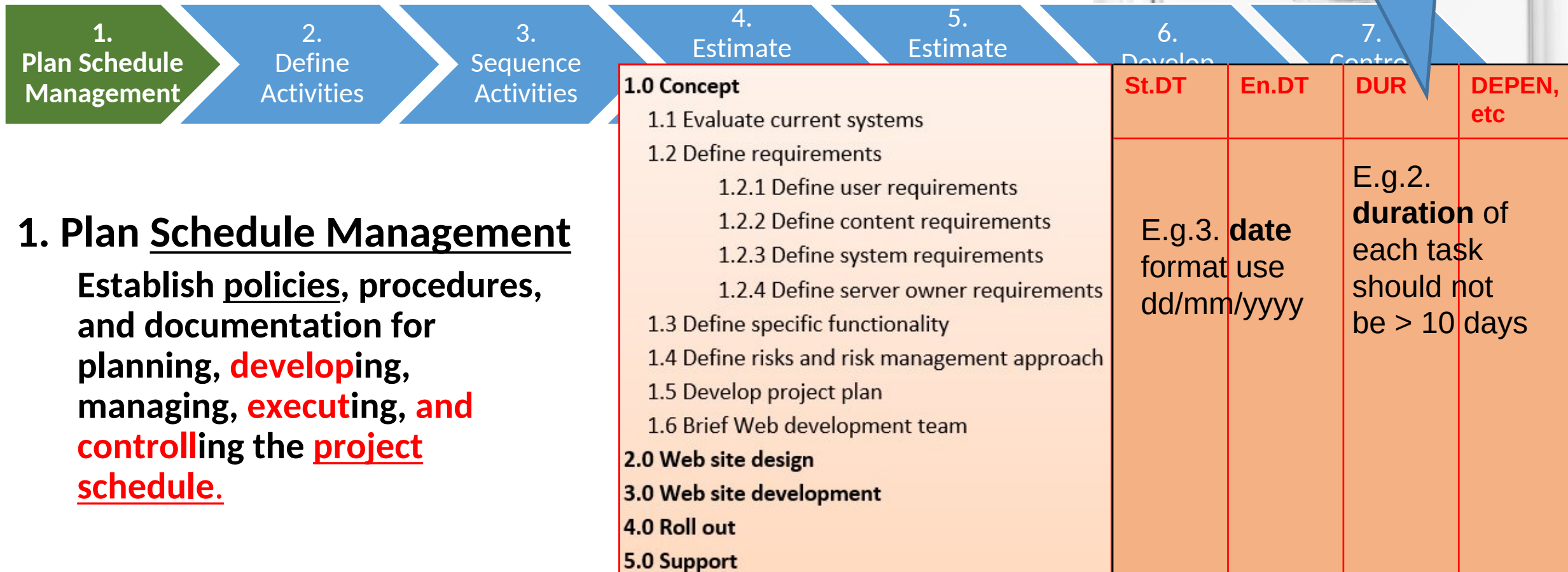
1. Plan Schedule Management
2. Define Activities
3. Sequence Activities
4. Estimate Activity Resources
5. Estimate Activity Durations
6. Develop Schedule
7. Control Schedule



Project Time Management

1.4 Knowledge & Attributes of a Project Manager

E.g1. unit of measurement
for duration is
(hours/days/weeks)

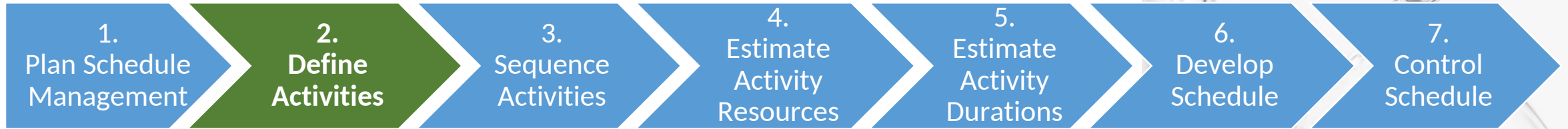


1. Plan Schedule Management

Establish policies, procedures, and documentation for planning, **developing**, managing, **executing**, and **controlling** the project schedule.

Project Time Management

1.4 Knowledge & Attributes of a Project Manager



2. Define Activities

- **Identify the specific activities (from the WBS) that the project team members and stakeholders must perform to produce the project deliverables.**
- **Outputs:** activity list, activity attributes, and milestone list.

Project stages

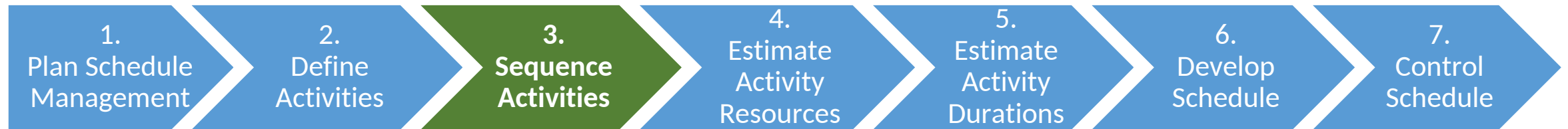
What are activity attributes?

Ans - detailed description of an activity.

Examples: description of the activity, responsible person, resources required, location, etc

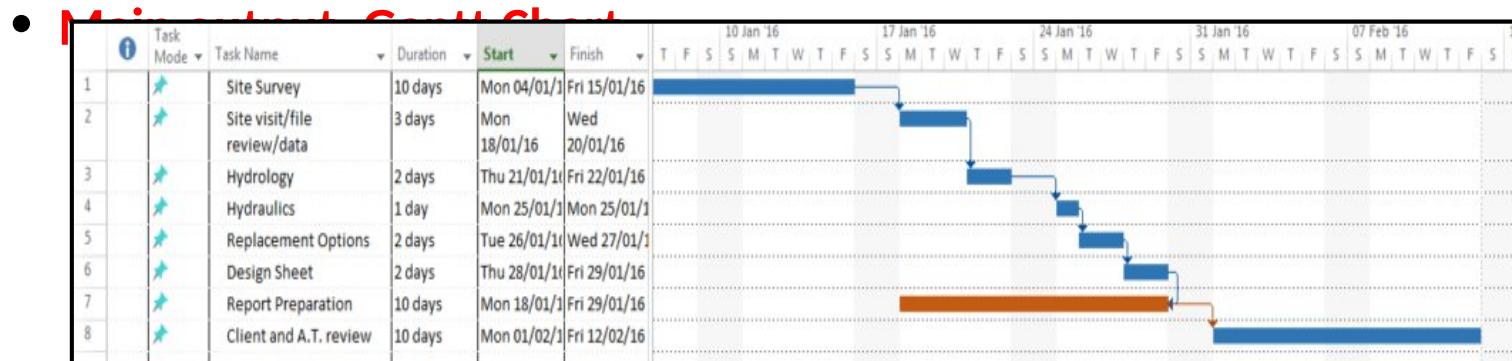
Project Time Management

1.4 Knowledge & Attributes of a Project Manager



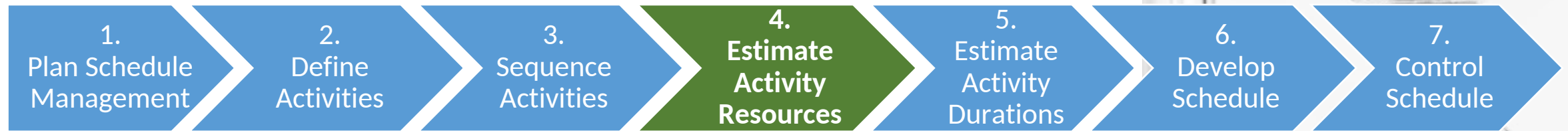
3. Sequence Activities

- Identify and document the relationships between project activities.
(specify the predecessor to sequence project activities)



Project Time Management

1.4 Knowledge & Attributes of a Project Manager

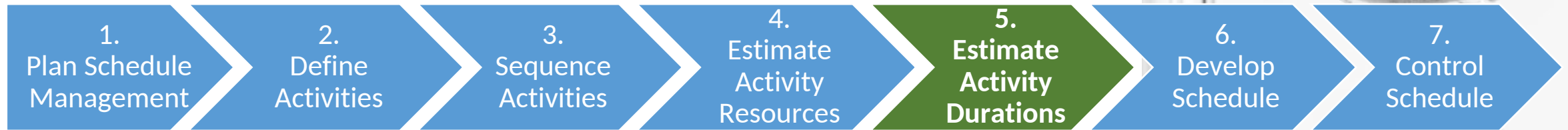


4. Estimate Activity Resources

- Estimate the amount of resources (**people, equipment, and materials**) needed to perform the activities.
- Outputs: activity resource requirements, resource breakdown structure

Project Time Management

1.4 Knowledge & Attributes of a Project Manager

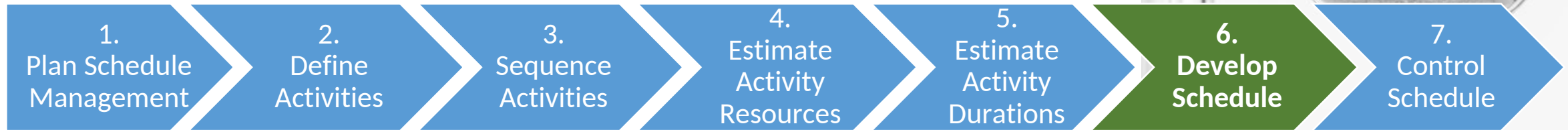


5. Estimate Activity Durations

- **Estimate durations(days/weeks) needed to complete each activity.**
- Tool and techniques: **three-point estimates**
- Main outputs: activity duration estimates.

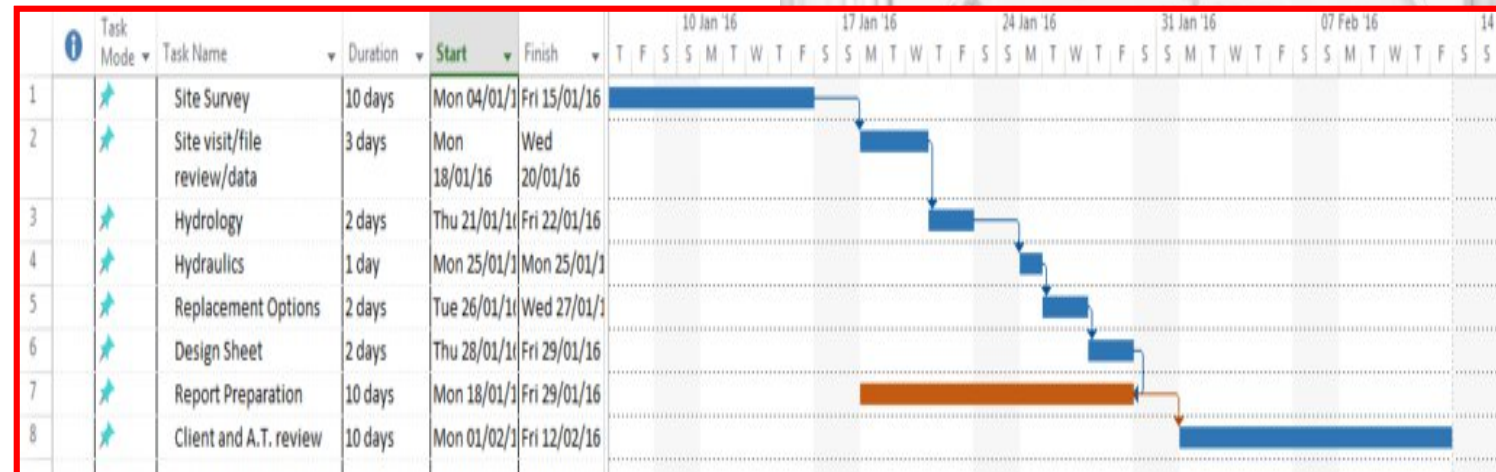
Project Time Management

1.4 Knowledge & Attributes of a Project Manager



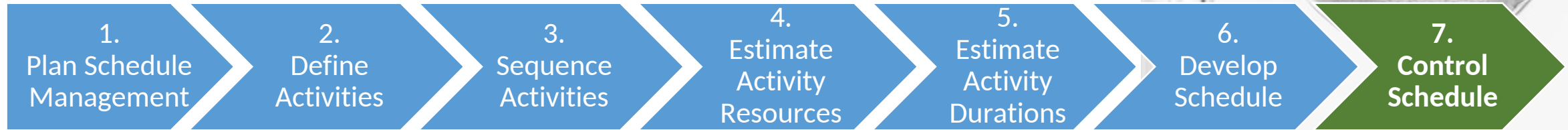
6. Develop Schedule

- Analyze activity sequences, activity resource estimates, and activity duration estimates to create the project schedule.
- Outputs:** project schedule, schedule baseline, schedule data, project calendars.



Project Time Management

1.4 Knowledge & Attributes of a Project Manager



7. Control Schedule

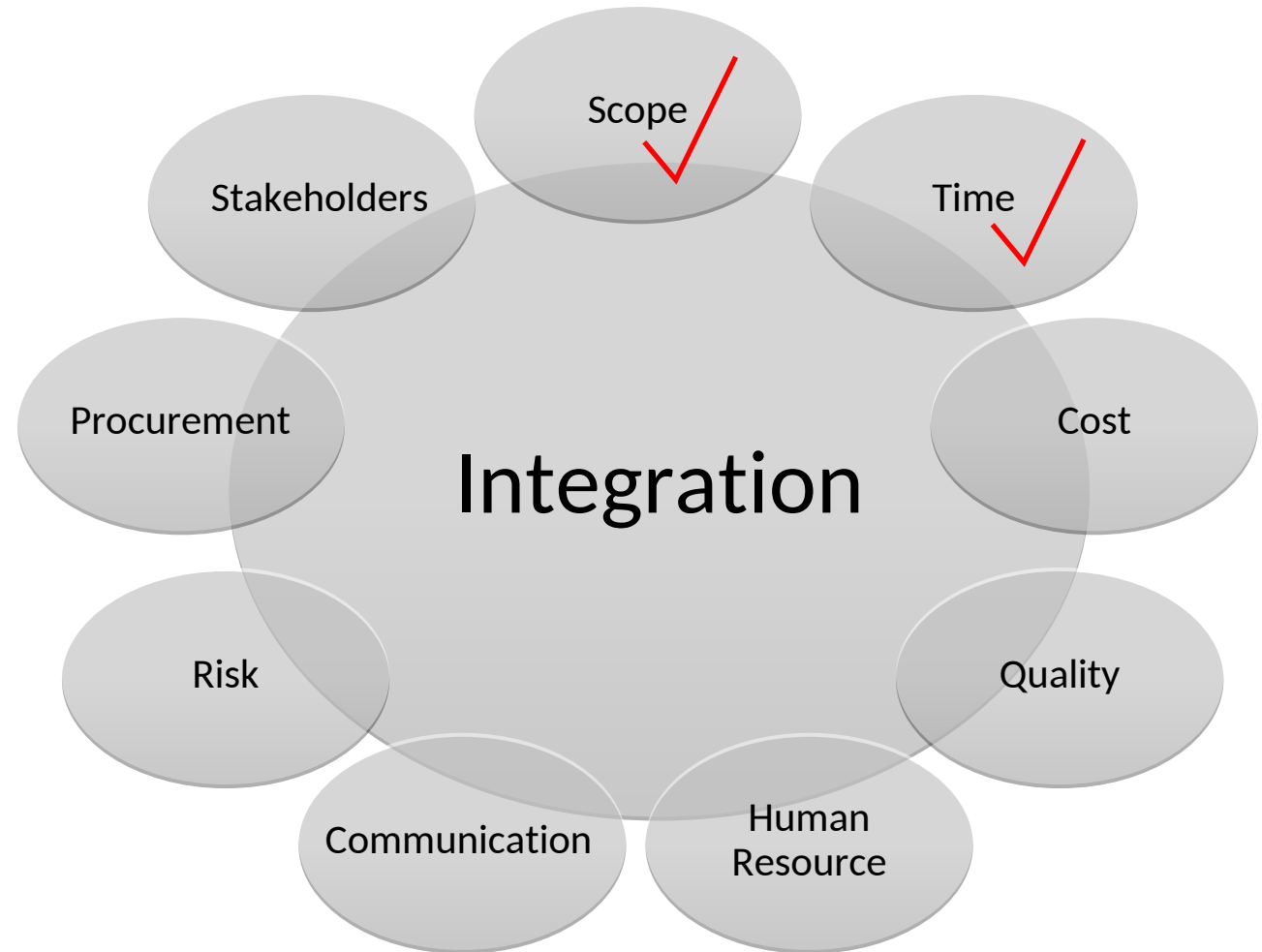
Monitor and manage changes to the project schedule to ensure the project is completed on time.

- **Update the schedule as needed** to reflect changes
- **Take corrective actions if there is any schedule issues.**

Knowledge of a PM

1.4 Knowledge & Attributes of a Project Manager

10 Knowledge Areas



A close-up photograph of a stack of several coins, likely Euro coins, resting on a financial document. The document features a table with dates and numbers, a ruler, and a calculator. The dates listed include 2011.06.30, 2011.09.30, 2011.12.31, 2012.03.31, 2012.06.30, and 2012.09.30. The numbers include 2 754 605, 2 814 605, 2 824 605, 19 301, 91 50, 3 0, 9 00, 8 000 000, and 7 000 000. The calculator shows the number 6 on its display.

Project Cost Management

1.4 Knowledge & Attributes of a Project Manager

- Includes the processes required to ensure that project team completes a project within an approved budget.
- 4 processes:
 1. Plan Cost Management
 2. Estimate Cost
 3. Determine Budget
 4. Control Costs



Project Cost Management

1.4 Knowledge & Attributes of a Project Manager



1. Plan Cost Management

- Establish the policies, procedures, and documentation for planning, managing, expending, and controlling project costs.
- Outputs: cost management plan.

Project Cost Management

1.4 Knowledge & Attributes of a Project Manager

2. Estimate Cost

- To **estimate** cost need to know all the resources needed to complete the project - E.g. **people, hardware, software**, etc
(mentioned also the quantity of each resource needed)

- Outputs:** activity cost estimates & basis of estimates

(The **assumptions** made when preparing the costs estimate)

Cost
item by
category

Surveyor Pro Project Cost Baseline Created October 10, 2008

WBS Items	1	2	3	4	5	6	7	8	9	10	11	12	Totals
1. Project Management													
Project manager	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	96,000
Project team members	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	144,000
Contractors		6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	66,300
2. Hardware													
2.1 Handheld devices				30,000	30,000								60,000
2.2 Servers				8,000	8,000								16,000
3. Software													
3.1 Licensed software				10,000	10,000								20,000
3.2 Software development		60,000	60,000	80,000	127,000	127,000	90,000	50,000					594,000
4. Testing			6,000	8,000	12,000	15,000	15,000	13,000					69,000
5. Training and Support													
Trainee cost									50,000				50,000
Travel cost									8,400				8,400
Project team members							24,000	24,000	24,000	24,000	24,000	24,000	144,000
6. Reserves				10,000	10,000	30,000	30,000	60,000	40,000	40,000	30,000	3,540	253,540
Totals	20,000	86,027	92,027	172,027	223,027	198,027	185,027	173,027	148,427	90,027	80,027	53,567	1,521,240

Project Cost Management

1.4 Knowledge & Attributes of a Project Manager

Surveyor Pro Project Cost Baseline Created October 10, 2008*

WBS Items	1	2	3	4	5	6	7	8	9	10	11	12	Totals
1. Project Management													
Project manager	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	96,000
Project team members	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	144,000
Contractors		6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	66,300
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Totals	20,000	86,027	92,027	172,027	223,027	198,027	185,027	173,027	148,427	90,027	80,027	53,567	1,521,240

COST = forecasted amount to pay to complete a project

3. Determine Budget (BUDGET ≈ spending plan (anticipated amount of \$ to spend from time to time to complete a project))

- Aggregate the estimated costs of individual activities or work packages to establish an authorized cost baseline.
- Main outputs: cost baseline(*approved budget*), project funding reqs



Project Cost Management

1.4 Knowledge & Attributes of a Project Manager



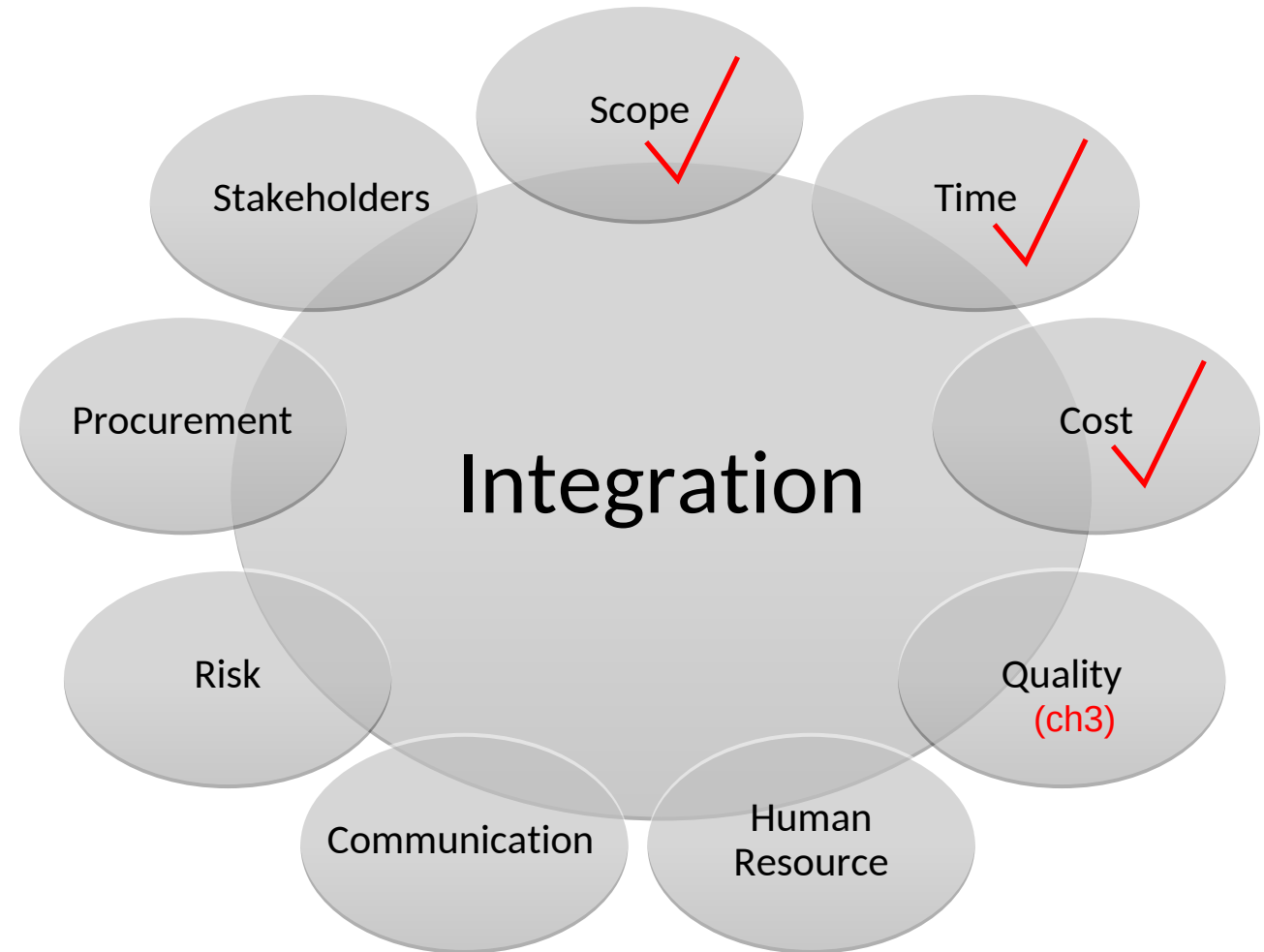
4. Control Costs

- Control changes to the project budget.
- Tool and Technique: **use Earned Value Management (EVM) technique to determine whether a project is**
 - Within budget
 - On time
- Main outputs: work performance measurements, cost forecasts.

Knowledge of a PM

1.4 Knowledge & Attributes of a Project Manager

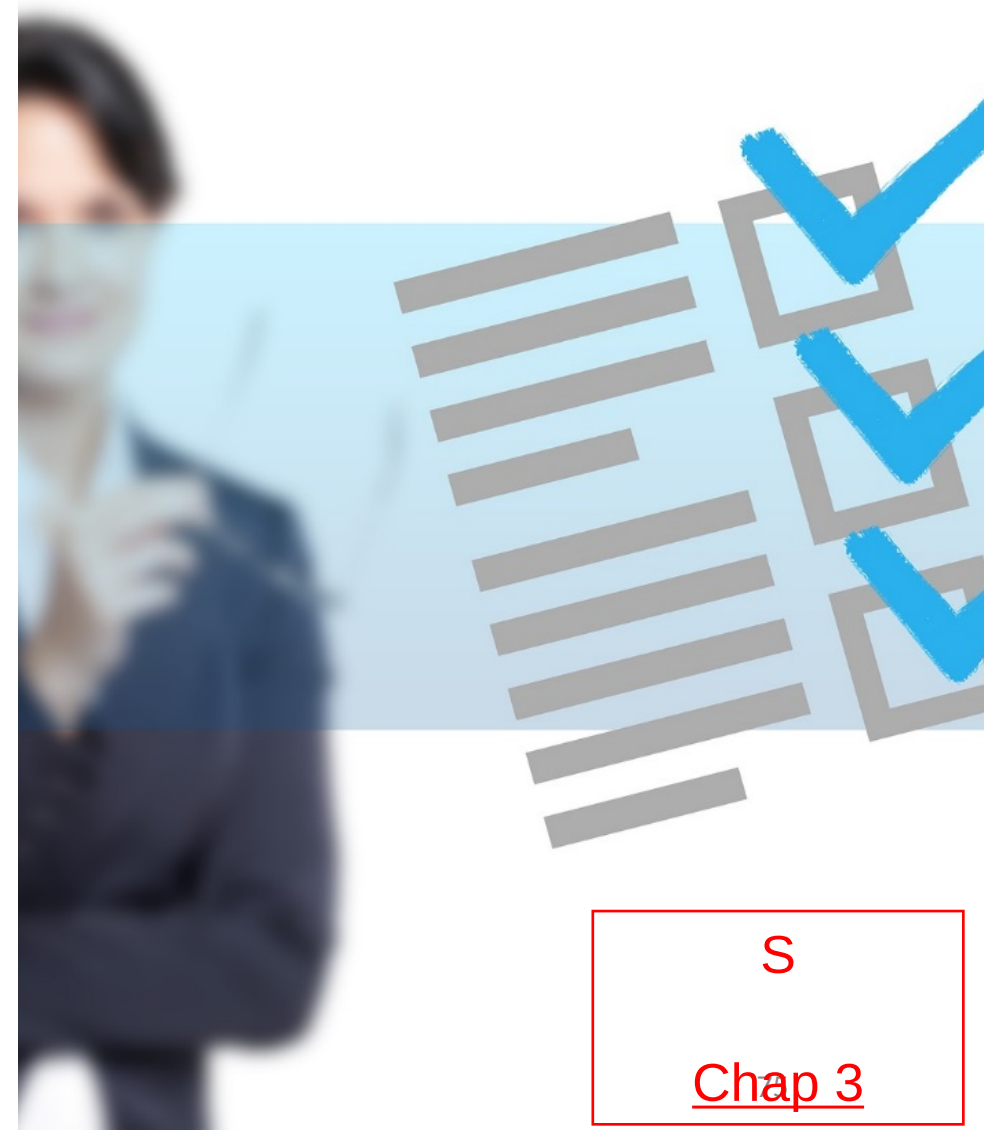
10 Knowledge Areas



Project Quality Management

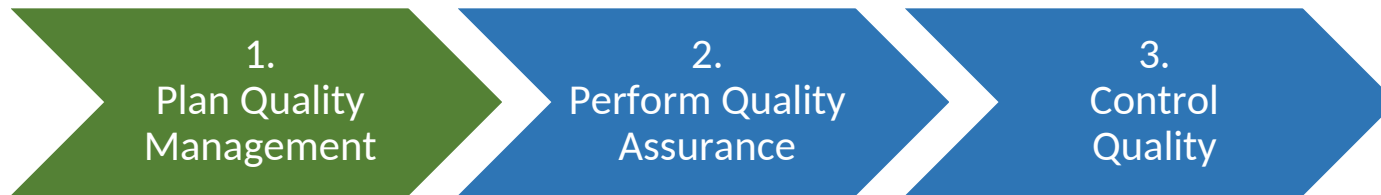
1.4 Knowledge & Attributes of a Project Manager

- To ensure that the project will satisfy the needs for which it was undertaken.
- 3 processes
 1. Plan Quality Management
 2. Perform Quality Assurance
 3. Control Quality



Project Quality Management

1.4 Knowledge & Attributes of a Project Manager



1. Plan Quality Management

- Identify which quality standards are relevant to the project and how to satisfy those standards.
- Quality standards for IT project, e.g. allow the system growth, reasonable response time, the system produces consistent and accurate information.
- Main outputs: quality management plan, quality metrics, quality checklists, process improvement plan.

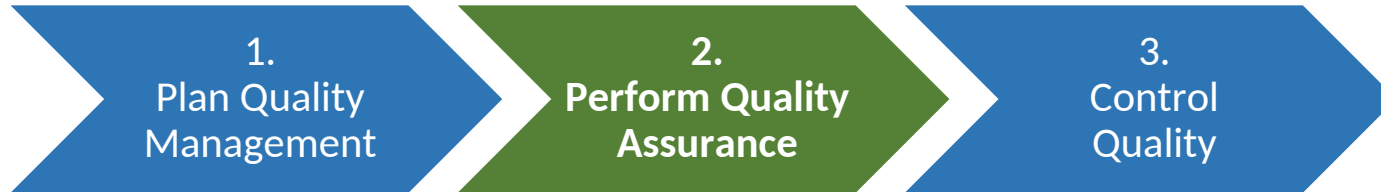
Note:

metric is a standard of measurement, e.g. failure rates of products produced, availability of goods and services, customer satisfaction ratings.

[Chap 3](#)

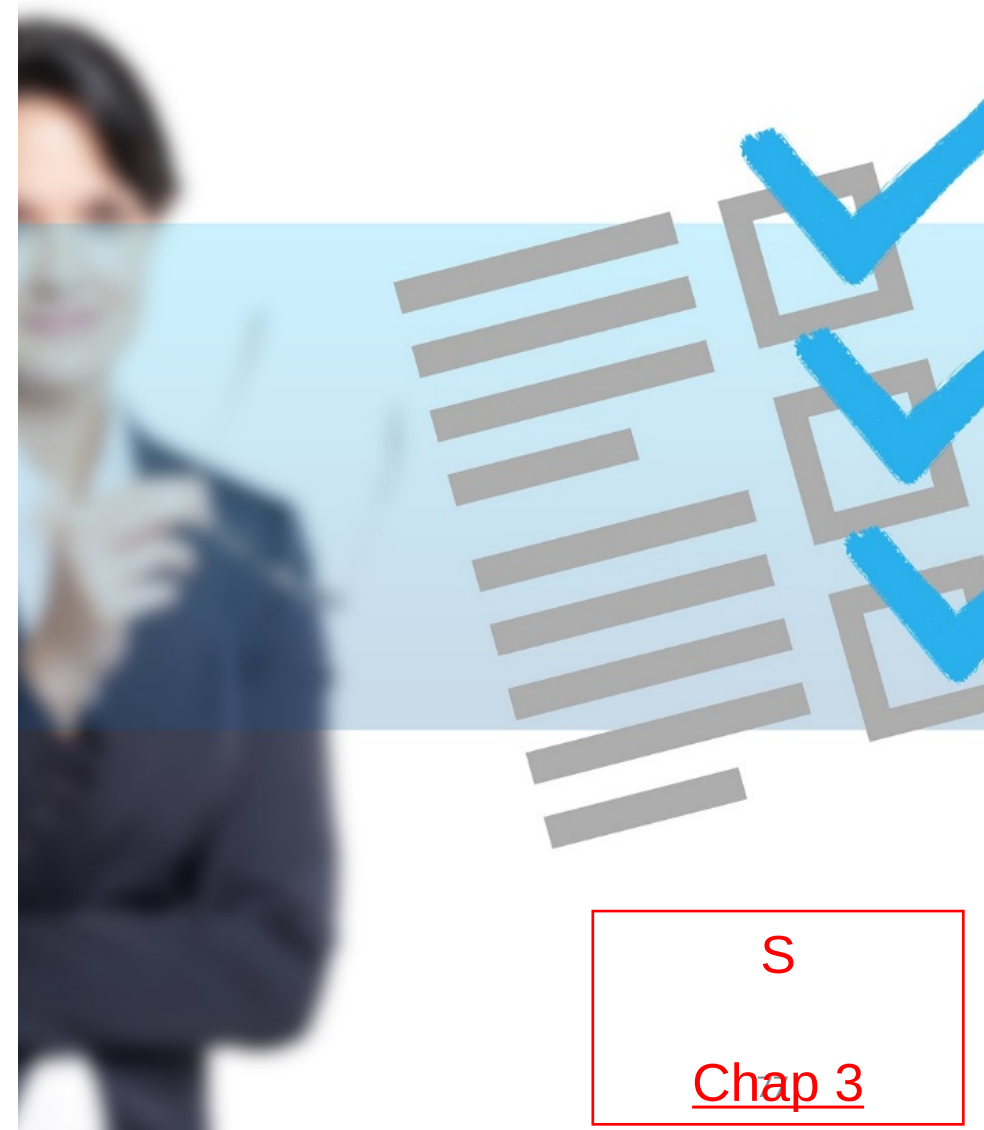
Project Quality Management

1.4 Knowledge & Attributes of a Project Manager



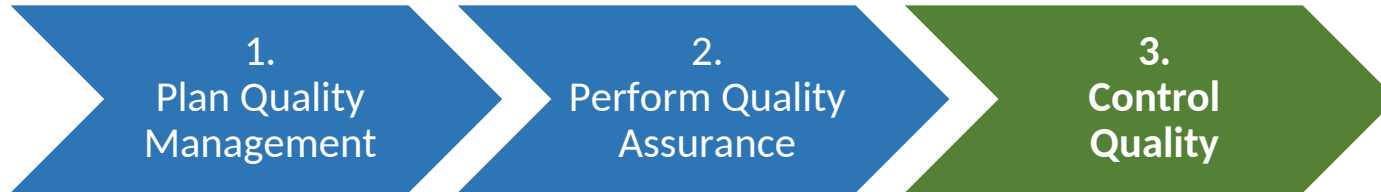
2. Perform Quality Assurance

- Periodically evaluate overall project performance to ensure that the project will satisfy the relevant quality standards.



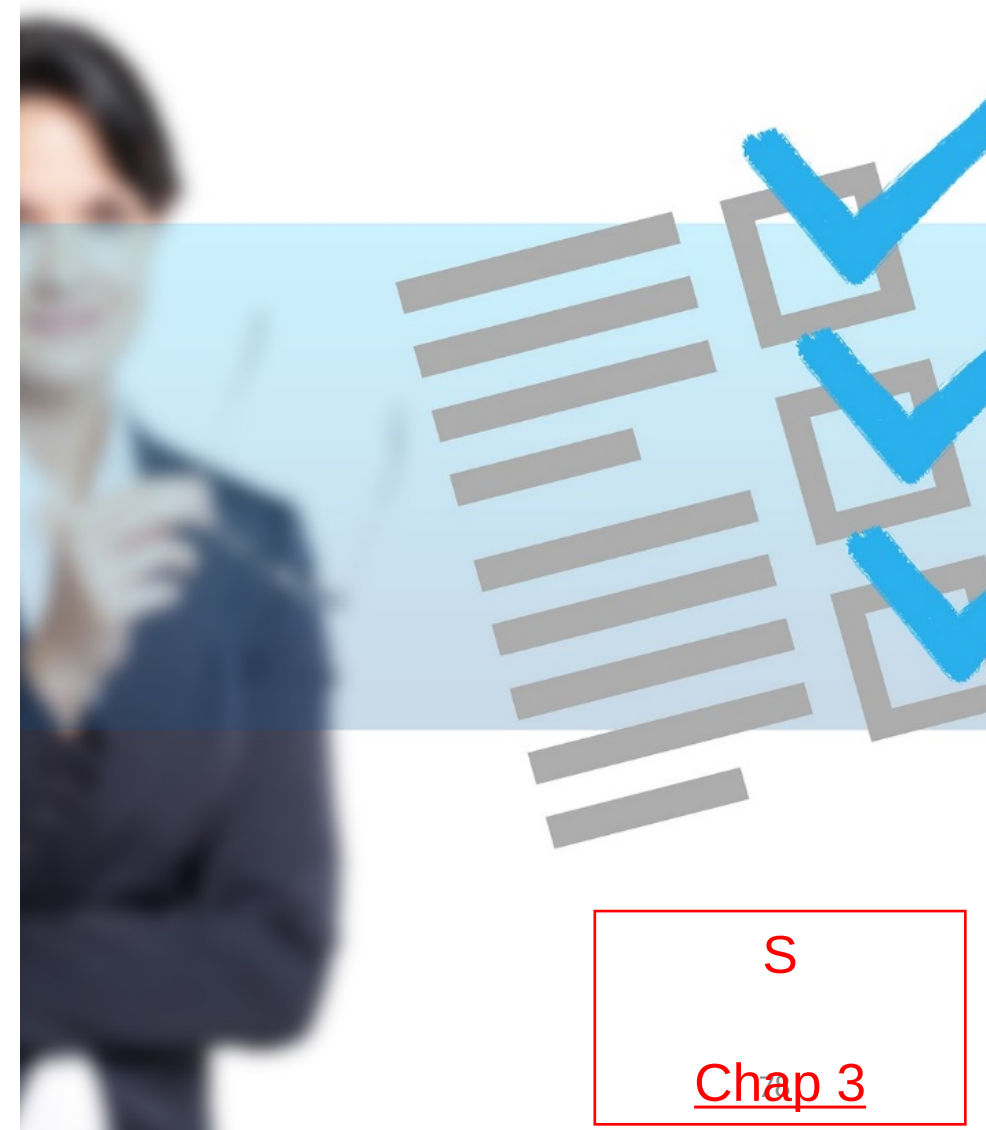
Project Quality Management

1.4 Knowledge & Attributes of a Project Manager



3. Control Quality

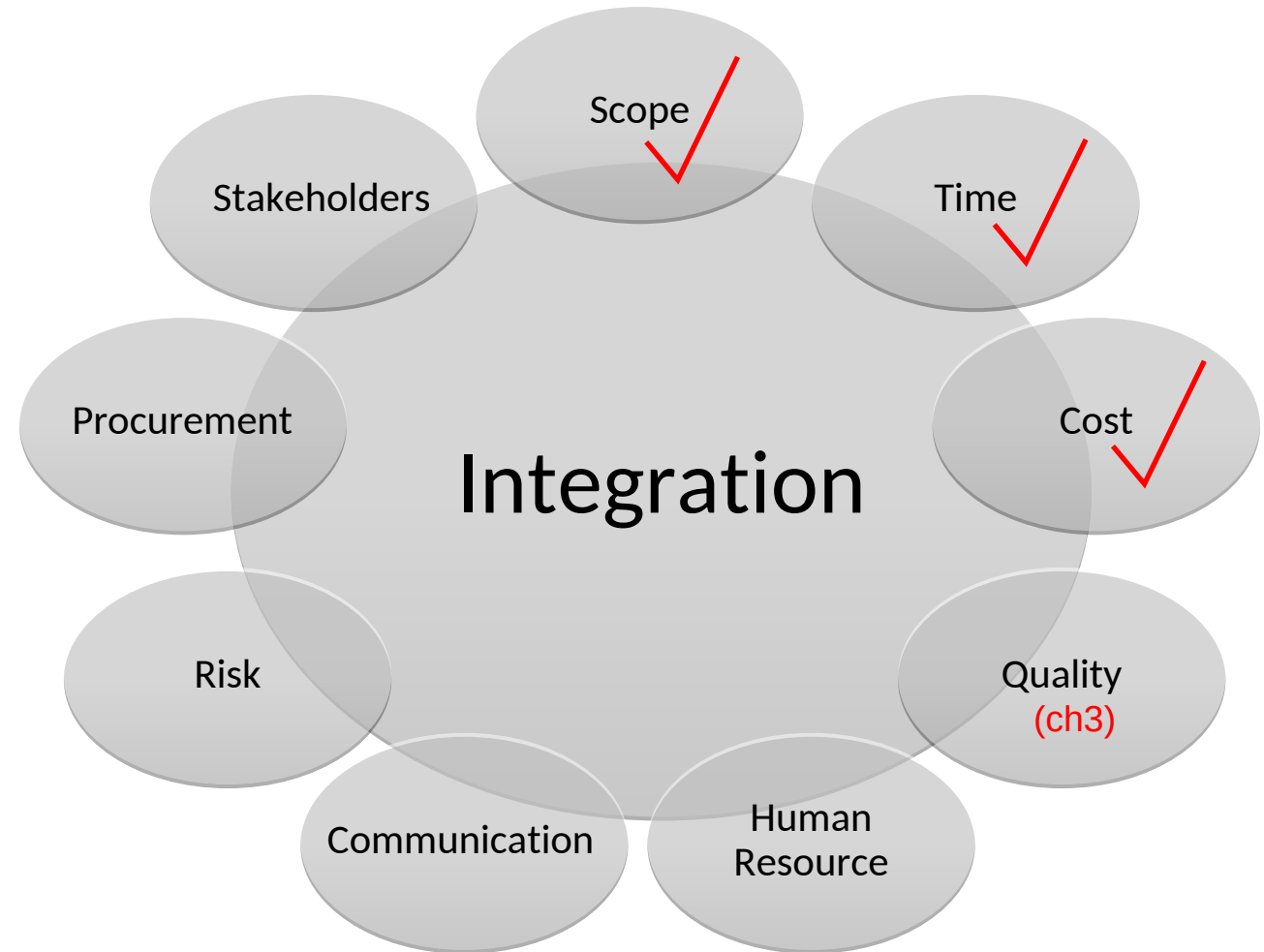
- Monitor and record results of executing the quality activities to assess performance and recommend necessary changes.



Knowledge of a PM

1.4 Knowledge & Attributes of a Project Manager

10 Knowledge Areas





Project Human Resource Management

1.4 Knowledge & Attributes of a Project Manager

- Includes the processes required to make the most effective use of the people involved in a project.
- Includes all project *stakeholders*: sponsors, customers, project team members, support staff, suppliers supporting the project, etc.
- 4 processes:
 1. Plan HR Management
 2. Acquire Project Team
 3. Develop Project Team
 4. Manage Project Team



Project Human Resource Management

1.4 Knowledge & Attributes of a Project Manager



1. Plan HR Management

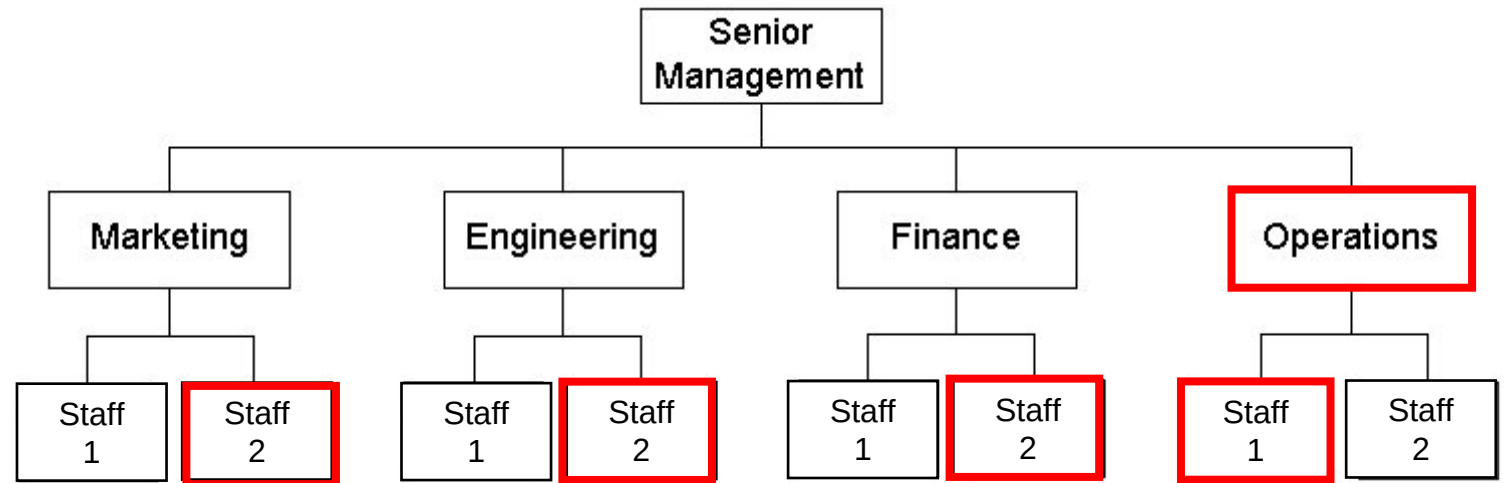
- Identify and document project resources, roles, responsibilities, and reporting relationships.
- Establish an **organization structure** for the project:
 - i. The Functional Structure
 - ii. The Project Structure
 - iii. The Matrix Structure
- **Output: HR plan**

Project Human Resource Management

1.4 Knowledge & Attributes of a Project Manager

Organization structures for managing projects

- (i) **Functional Organizational Structure**
(using an existing organization structure to manage Project)



E.g. Transfer equipment to new office

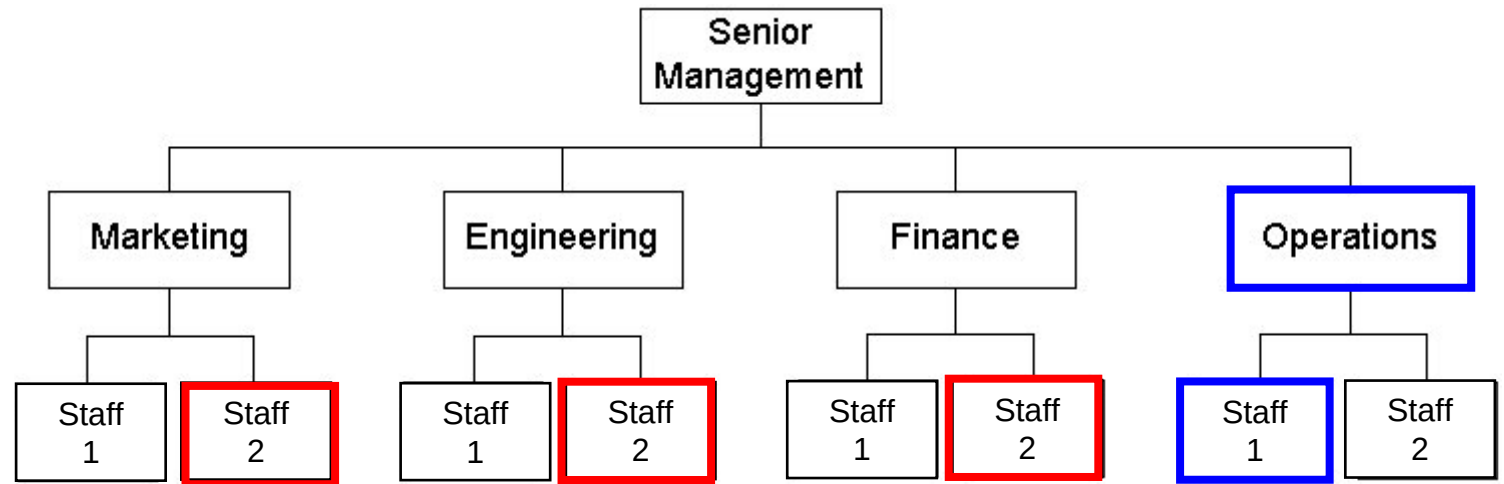
This project is managed by the **Operations manager** and assisted by **ONE** personnel from every other departments

Project Human Resource Management

1.4 Knowledge & Attributes of a Project Manager

Organization structures for managing projects

(i) Functional Organizational Structure



Pros

- Cost savings

Cons

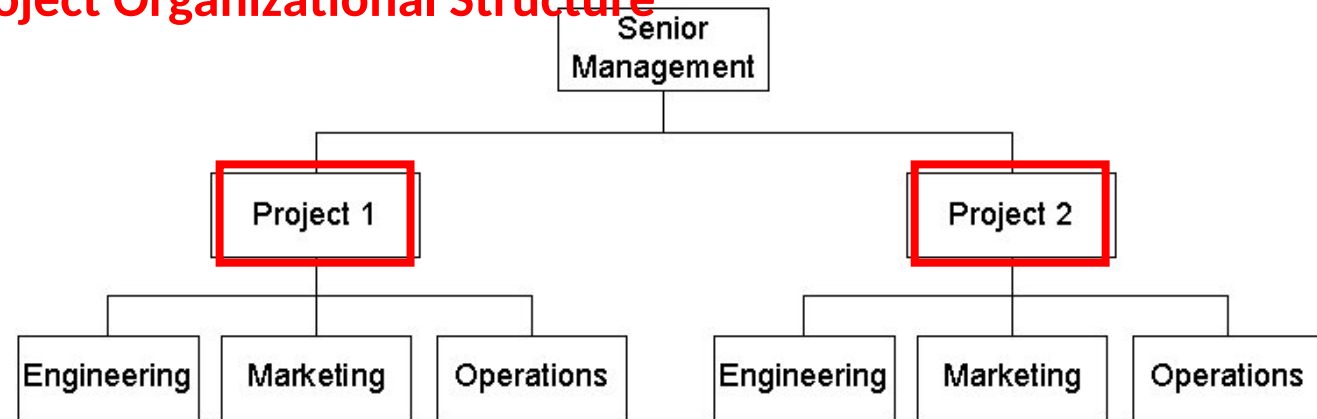
- Project leaders have little power
- Conflicts of priorities (*Transfer equipment to new office is not the staffs regular jobs*)

Project Human Resource Management

1.4 Knowledge & Attributes of a Project Manager

Organization structures for managing projects

(ii) Project Organizational Structure



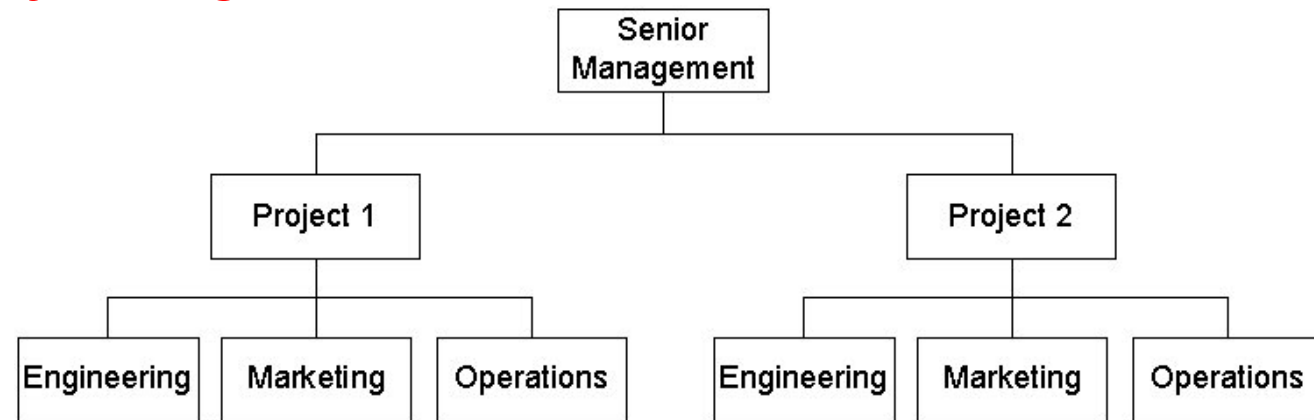
- Project managers report to the **CEO / Sr Mgmt**
- E.g. of **businesses** where work is organized around projects:
 - **Software development companies**
 - **Construction companies**
 - **Event mgmt. companies**

Project Human Resource Management

1.4 Knowledge & Attributes of a Project Manager

Organization structures for managing projects

(ii) Project Organizational Structure



Advantages

- Unity of command
(each team member report to and receive order form only one manager)
- Highly responsive to project's objectives/ customer needs

Disadvantages

- Duplication of facilities
- Cost inefficiency of resources

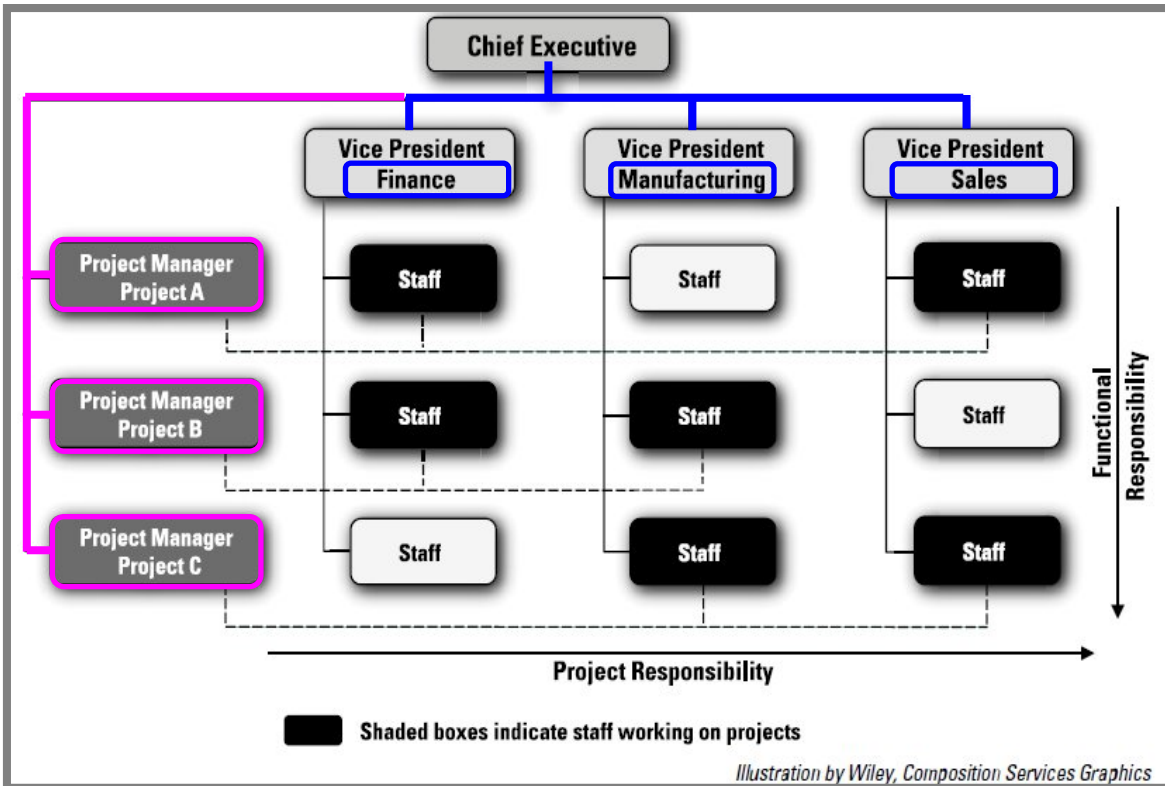
Project Human Resource Management

1.4 Knowledge & Attributes of a Project Manager

Organization structures for managing projects

(iii) Matrix Organizational Structure

- Combination of **functional** and **project structures**.
- **Project managers often have staffs from various functional areas** working on their projects. (E.g Project C _____)
- **Adv:** efficient use of resources
- **Disadv:** one staff report to 2 managers/bosses





Project Human Resource Management

1.4 Knowledge & Attributes of a Project Manager



Quick Review ...

3 Organization structures for the project:

- i. The Functional Structure
- ii. The Project Structure
- iii. Matrix Structure



Project Human Resource Management

1.4 Knowledge & Attributes of a Project Manager



2. Acquire Project Team

- Get the personnel needed and assigned them to work on the project
- **Tool** (e.g. Microsoft Project) and techniques:
 - a. **Resource assignment**
(to specify who perform what task)
 - b. **Resource loading**
(analyse of the total amt of work assigned to a resource over a time period)
 - c. **Resource leveling**
- **Outputs**: project staff assignments, resource calendars.

Tool & techniques for project team acquisition

```
graph TD; A[Tool & techniques for project team acquisition] --> B[Resource assignment]; A --> C[Resource loading]; A --> D[Resource leveling];
```

Resource assignment

To assign particular personnel to their projects or to acquire additional human resources needed to staff the project.

- PM with strong influencing & negotiating skills are often good at getting internal people to work on their projects.
- Main outputs: project staff assignments, resource availability information, updates to the staffing management plan, project team directory.

Resource loading

- Amount of individual resources an existing schedule requires during specific time periods.
- Help PM and individuals to develop schedules.
- Overallocation: more resources than are available are assigned to perform work at a given time. E.g. individual have to work 24 hours a day to meet the allocated schedule.

Resource leveling

- A technique for resolving resource conflicts (overallocation & underallocation).
- Purpose: to create a smoother distribution of resource usage.
- PM examine the PERT chart for areas of slack or float, and to identify resource conflicts.
- E.g. remove overallocations by delaying noncritical tasks, which does not result in an overall schedule delay.
- Therefore, resource leveling aims to minimize period-by-period variations in resource loading by shifting tasks within their slack allowances.

Project Human Resource Management

1.4 Knowledge & Attributes of a Project Manager



3. Develop Project Team

- **Purpose:** to ensure that people can work together more effectively to improve project performance.
- Main outputs: team performance assessments, enterprise environmental factors updates.
- **The Tuckman 5 stages of team development:**
 - a. Forming (*first stage*)
 - b. Storming
 - c. Norming
 - d. Performing

Describes what a team will go through from start to end.

The Tuckman's 5 stages of Team Development

1.4.1 e. Project HR Management

Forming

- Project team is form and **members are introduced to each other**
(e.g. a team consisting of developers, testers and a UI designer is formed for a new mobile app project)
- This stage is necessary but **little work is achieved**

Storming

- **Conflict emerges** as team members have differing opinions, challenge ideas, and compete for influence.
- This is a turbulent stage where **disagreements** occur
- (e.g. Developers start writing code, but disagreements arise over coding standards (indentation, naming conventions, percentage of comment lines, etc.)

Norming

- This stage is achieved when **conflicts are resolved**
(established acceptable rules and standards)
- **Team members begins to collaborate effectively**

Performing

- **Project team is united.** Problems are brought up, discussed and then solved
- Level of **work performance is high**; Productivity at its optimum

Adjourning

- **The team is dissolved** after all works have been successfully completed and reaches its goals.



Project Human Resource Management

1.4 Knowledge & Attributes of a Project Manager



4. Manage Project Team:

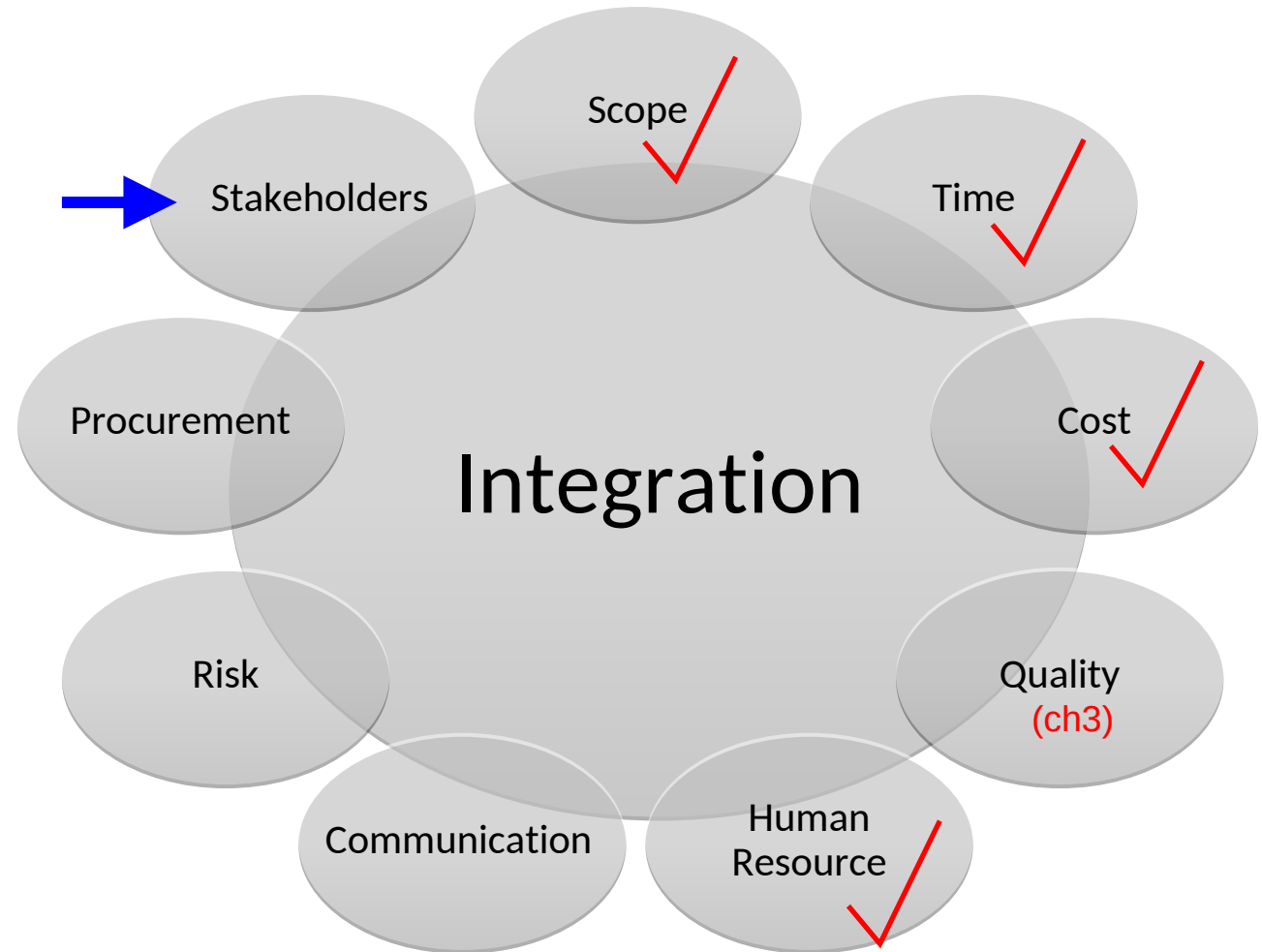
- Track team member performance,
- Motivate team members,
- Provide timely feedback,
- Resolve issues and conflicts
- etc

PM must ensure that people can work together as a team to achieve project goals

Knowledge of a PM

1.4 Knowledge & Attributes of a Project Manager

10 Knowledge Areas





Project Stakeholder Management

1.4 Knowledge & Attributes of a Project Manager

- **Purpose:** to identify the people, groups or organizations that could impact or be impacted by the project,
- to analyze stakeholder expectations and their impact on the project
- to develop appropriate management strategies to effectively engage stakeholders in project decisions and execution.
- **4 processes:**
 1. Identify Stakeholders
 2. Plan Stakeholder Engagement
 3. Manage Stakeholder Engagement
 4. Monitor Stakeholder Engagement



Project Stakeholder Management

1.4 Knowledge & Attributes of a Project Manager



1. Identify Stakeholders

- **Identify the people, groups, or organizations that could impact or be impacted by a decision, activity, or outcome of the project;** and analyze and document relevant information regarding their interests, involvement, interdependencies, influence, and potential impact on project success.
- **Output:** stakeholder register.





Project Stakeholder Management

1.4 Knowledge & Attributes of a Project Manager

Stakeholder register (e.g. 1)

(Project = To develop a **Web-based Attendance System (WAS)** for AAA University)

No	Indiv/GROUP	JOB POSITION	ROLE	Expectation
1	Top management	President, Vice presidents	Project sponsor	
2	Academics	Professors, lecturers, tutors.	Users (generate attendance code for students)	
3	Students	Course-rep, class-rep, ordinary students	Users (taking own class attendance)	
4	JACK	Project manager	Manage the WAS development project	
5	Team members	Developers, testers	Develop & test WAS	
6	...			
7	...			

The people, groups or organizations that could impact or be impacted by a decision, activity or outcome of the project

Stakeholder Management

If you are assigned to manage a project, you need to know who are the stakeholders

Knowledge & Attributes of a Project Manager

Stakeholder register (e.g. 2)

(Project = To develop a **Web-based Attendance System (WAS)** for AAA University)

No	Indiv/GROUP	JOB POSITION	ROLE	Expectation
1	Top management	President, Vice presidents, Deans	Project sponsor	
2	Academics	Professors, lecturers, tutors.	Users (generate attendance code for students, ...)	
3	Students	Course-rep, class-rep, ordinary students	Users (taking own class attendance, ...)	
4	JACK	Project manager	Manage the WAS development project	
5	JIM	UI & DB Designer	Design sys UI & Database	
6	JOHN	Developer	Develop & test the WAS	
7	...			

Work with **stakeholders** to meet their needs/expectations



Project Stakeholder Management

1.4 Knowledge & Attributes of a Project Manager

1. Identify Stakeholders

Identify the people, groups, or organizations that could impact or be impacted by the project; and analyze and document relevant information regarding their interests, interdependencies, influence, and potential impact on project success.

PS: A high-ranking manager who champions and supports a project
(Larson, 2021)

Output: **Stakeholder register (e.g., 2)**

Name	Position	Internal/ External	Project Role	Contact Information
Stephen	VP of Operations	Internal	Project sponsor	stephen@globaloil.com
Betsy	CFO	Internal	Senior manager, approves funds	betsy@globaloil.com
Chien	CIO	Internal	Senior manager, PM's boss	chien@globaloil.com
Ryan	IT analyst	Internal	Team member	ryan@globaloil.com
Lori	Director, Accounting	Internal	Senior manager	lori@globaloil.com
Sanjay	Director, Refineries	Internal	Senior manager of largest refinery	sanjay@globaloil.com
Debra	Consultant	External	Project manager	debra@gmail.com
Suppliers	Suppliers	External	Supply software	suppliers@gmail.com

(Schwalbe,
2019)



Project Stakeholder Management

1.4 Knowledge & Attributes of a Project Manager



2. Plan Stakeholder Management

- Develop appropriate management strategies to effectively **engage stakeholders** throughout the project life cycle, based on the analysis of their needs, interests, and potential impact on project success.
- Output: stakeholder management plan.





Project Stakeholder Management

2. Plan Stakeholder Management

Develop appropriate management strategies to effectively engage stakeholders throughout the project life cycle, based on the analysis of their needs, interests, and potential impact on project success.

Output: stakeholder management plan (e.g.)

“Leading” status means Stephen is an **influential** person, who is also **actively** involved in the project

Name	Power/ Interest	Current Engagement	<u>Potential Management Strategies</u>
Stephen	High/high	Leading	Stephen can seem intimidating due to his physical stature and deep voice, but he has a great personality and sense of humor. He previously led a similar refinery upgrade program at another company and knows what he wants. <u>Manage closely and ask for his advice as needed. He likes short, frequent updates in person.</u>
Chien	High/ medium	Resistant	Chien is a very organized yet hardheaded man. He has been pushing corporate IT standards, and the system the PM and sponsor (Debra and Stephen) like best goes against those standards, even though it's the best solution for this project and the company as a whole. Need to convince him that this is okay and that people still respect his work and position.
Ryan	Medium/ high	Supportive	Ryan has been with the company for several years and is well respected, but he feels threatened by Debra. He also resents her getting paid more than he does. He wants to please his boss, Chien, first and foremost. Need to convince him that the suggested solution is in everyone's best interest.

Knows what he wants
and what he likes



Project Stakeholder Management

1.4 Knowledge & Attributes of a Project Manager



3. Manage Stakeholder Engagement/ relation

- Communicate and work with **stakeholders** to meet their needs/expectations, address issues as they occur, and foster appropriate stakeholder engagement in project activities throughout the project life cycle.
- Output: issue log, change requests.





Project Stakeholder Management

1.4 Knowledge & Attributes of a Project Manager



4. Monitor Stakeholder Engagement

- **Monitor stakeholder relationships and adjust plans and strategies to engage stakeholders as needed.**
- Main output: work performance information & change request

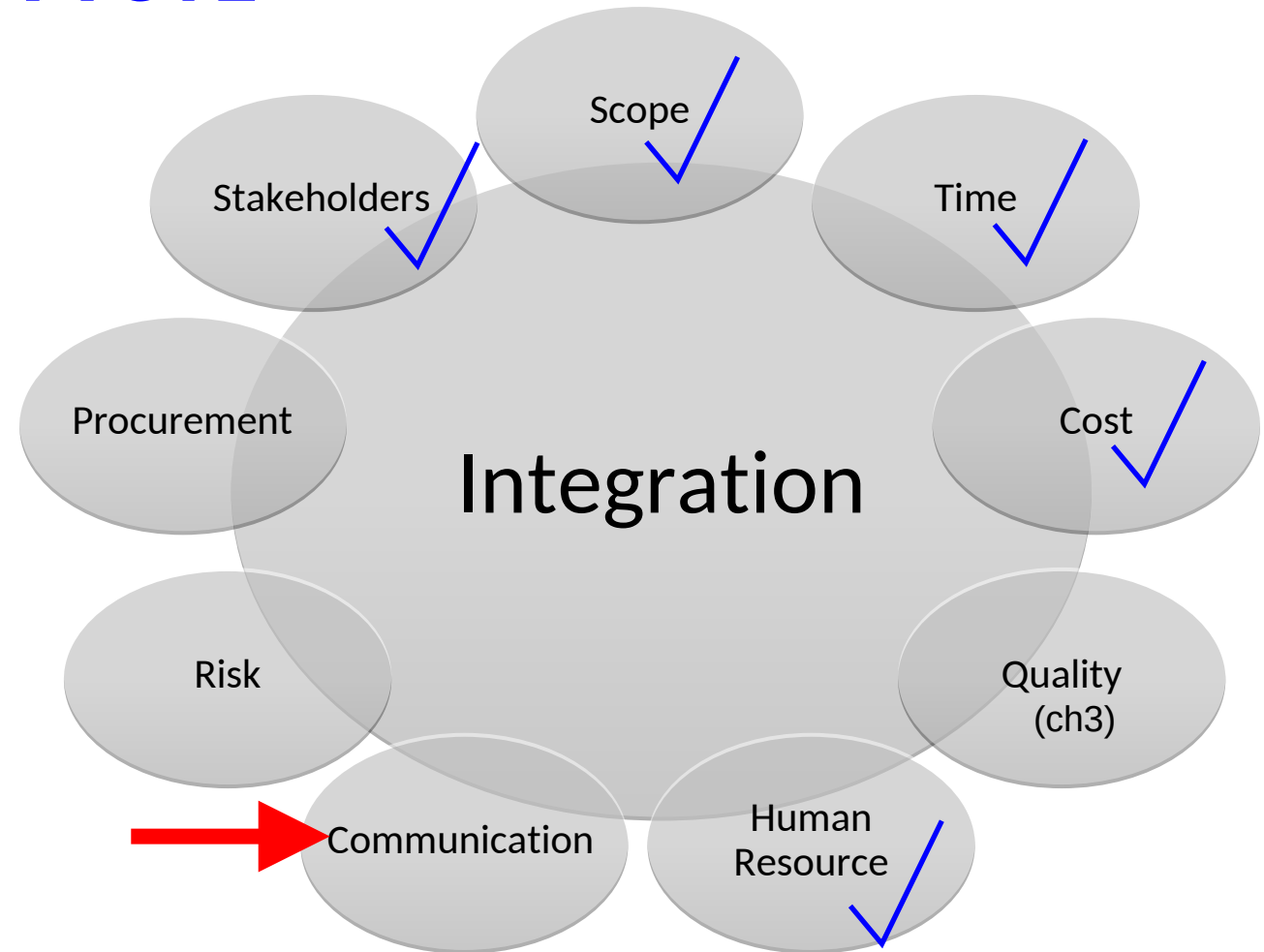


Knowledge of a PM

1.4 Knowledge & Attributes of a Project Manager

L3: Quick Review of Prevl

10 Knowledge Areas



Project Communication Management

1.4 Knowledge & Attributes of a Project Manager

- **Purpose:** To collect, disseminate and store of project information in a timely manner
- **3 processes:**
 1. Plan Communications Management
 2. Manage Communications
 3. Control communications



Project Communication Management

1.4 Knowledge & Attributes of a Project Manager

1. Plan Communication Management



- Develop an appropriate approach and **plan for project communications** based on stakeholder’s information needs and requirements, and available organizational assets.
- Key benefit: identifies and documents the approach to communicate most effectively and efficiently with stakeholders.

How to develop a Project Communication Plan?

PTM is to collect & disseminate project information

Description	Frequency	Method	Audience	Owner
Name of the communication	How often it will happen	Method of communication	Who will receive the communication	Who is responsible
Project team meeting	Daily	Meeting	Project team	Project manager
Stakeholder update	Monthly	Email newsletter	Stakeholders	Project manager
Board meeting	Every two weeks	Meeting	Project board	Project manager
Contribution to department newsletter	Quarterly	Section of newsletter	Wider development	Project manager to deliver to department administration

email newsletter to keep stakeholders informed of project progress

Project Communication Management

1.4 Knowledge & Attributes of a Project Manager

1. Plan Communications Management 2. Manage Communications 3. Control communications				
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2. Manage Communications

- Create, **collect, disseminate, store**, retrieve, and ultimately dispose **project information**

Project team (Developers, QA team)

Key benefit: enables an efficient and effective communications

SW Dev. Co. side (e.g CEO, PM)
Client side (e.g CEO, Dept mgrs, etc)

Means other relevant parties

Project Communication Management

1.4 Knowledge & Attributes of a Project Manager



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Contribution to department newsletter	Quarterly			

The purpose of COMM MGMT. is to collect, disseminate and store of project information in a timely manner so that stakeholders don't have to guess the STATUS of a project.

3. Control Communications

- Monitor and control communications throughout the entire project life cycle **to ensure the information needs of project stakeholders are met.**

(note: if some project stakeholders are no longer relevant to the project then these people should be removed from the audience list)

any moment in time.



Project Risk Management

1.4 Knowledge & Attributes of a Project Manager

- 6 major processes:
 1. Plan Risk Management
 2. Identify Risk
 3. Perform Qualitative Risk Analysis
 4. Perform Quantitative Risk Analysis
 5. Plan Risk Response
 6. Control Risk

Project Risk Management

1.4 Knowledge & Attributes of a Project Manager



1. Plan Risk Management

- Decide how to approach and plan the risk management activities for the project.
- By reviewing the project scope statement; cost, schedule, and communications management plans; enterprise environmental factors; and organizational process assets
- Main output: risk management plan

Project Risk Management

1.4 Knowledge & Attributes of a Project Manager



2. Identify Risk

- Determine which risks are likely to affect a project and documenting the characteristics of each
- Main output: the start of a risk register

Project Risk Management

1.4 Knowledge & Attributes of a Project Manager



3. Perform Qualitative Risk Analysis

- Prioritize risks based on their probability and impact of occurrence.
- After identifying risks, various tools and techniques can be used by project teams to rank risks and update information in the risk register
- Main output: risk register updates.

Project Risk Management

1.4 Knowledge & Attributes of a Project Manager



4. Perform Quantitative Risk Analysis

- Numerically estimate the effects of risks on project objectives.
- Main output: risk register updates.

Project Risk Management

1.4 Knowledge & Attributes of a Project Manager



5. Plan Risk Responses

- Take steps to enhance opportunities and reduce threats to meeting project objectives.
- Using outputs from the preceding risk management processes, project teams can develop risk response strategies that often result in updates to the risk register, project management plan, and other project documents as well as risk-related contract decisions

Project Risk Management

1.4 Knowledge & Attributes of a Project Manager

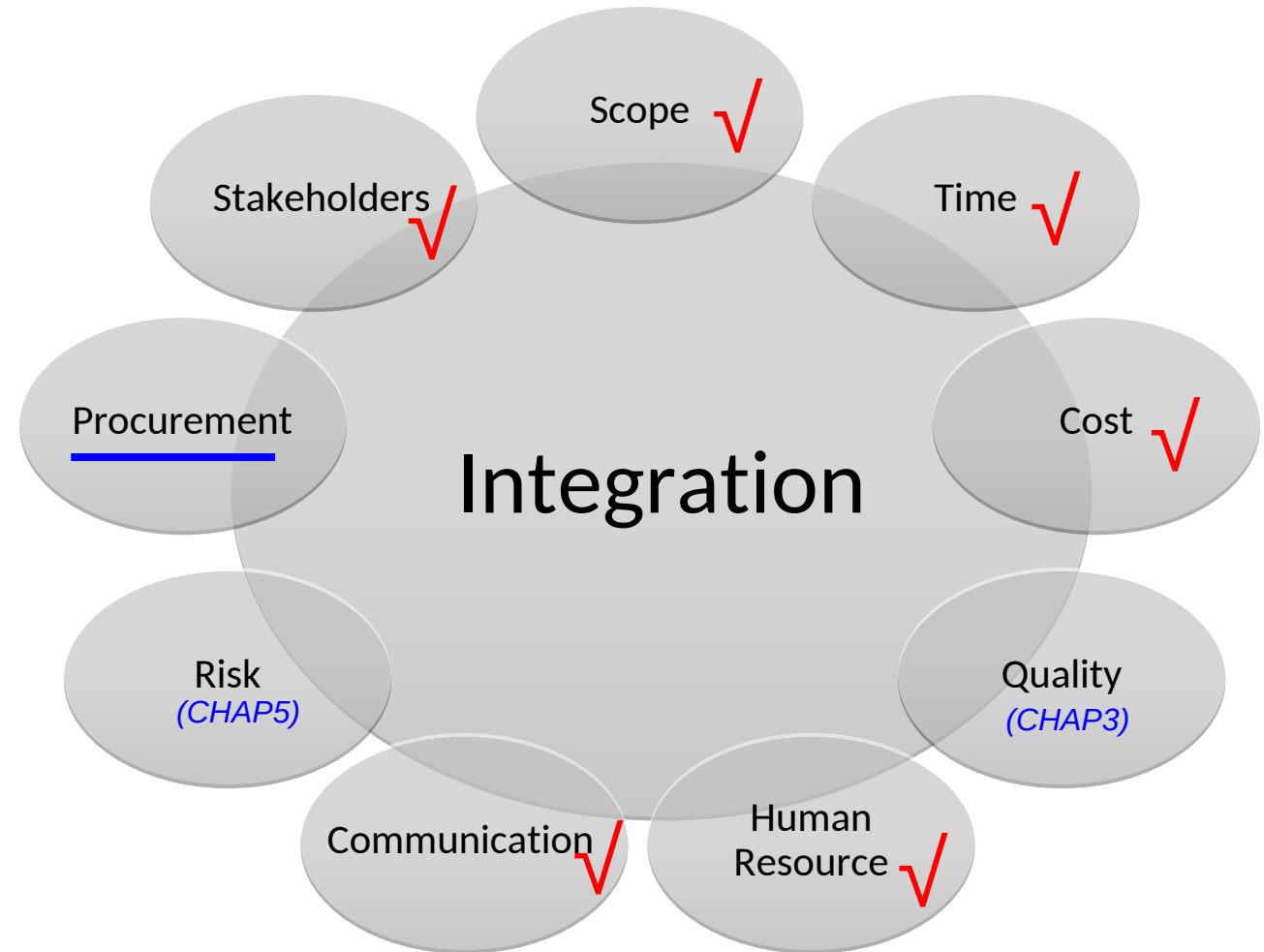


6. Control Risk

- Monitor identified and residual risks, identify new risks, carry out risk response plans, and evaluate the effectiveness of risk strategies throughout the life of the project.
- Main outputs: change requests and updates to the risk register.

1.4 Knowledge & Attributes of a Project Manager

10 Knowledge Areas



Project Procurement Management

1.4 Knowledge & Attributes of a Project Manager

- Includes the processes required **to acquire goods and services for a project from an outside source** (*supplier/vendor/contractor/seller*).
- Organizations can be either the buyer or seller of products or services under a contract.
- **3 main processes in Procurement Management:**
 1. Plan Procurement Management
 2. Conduct Procurement
 3. Control Procurement

Examples of Goods to acquire

Hardware:

servers, computers, laptops, tablets, mobile devices, printers, scanners, etc.

Software:

operating systems, development tools, project management software, database management systems, MS Office, etc

Examples of services to acquire

Software testing:

usability testing, etc

Training:

provide training for end-users

Project Procurement Management

1.4 Knowledge & Attributes of a Project Manager

Procurement Management Processes



1. Plan Procurement Management involve:

- Decide what & when to **ACQUIRE**, determine the type of contract, and describe the work for the *contractors/suppliers*.
- Outputs: procurement management plan, statements of work, make-or-buy decisions, procurement documents, source selection criteria, and change requests.



Project Procurement Management

1.4 Knowledge & Attributes of a Project Manager



2. Conduct Procurement

- **Invite *contractors/suppliers*** to submit proposal
- **Obtain *contractors/suppliers*** responses,
- **Select *contractors/suppliers***,
- **Award** contracts.
- Main output: Selected seller(s), agreement, change request



Project Procurement Management

1.4 Knowledge & Attributes of a Project Manager



3. Control Procurement

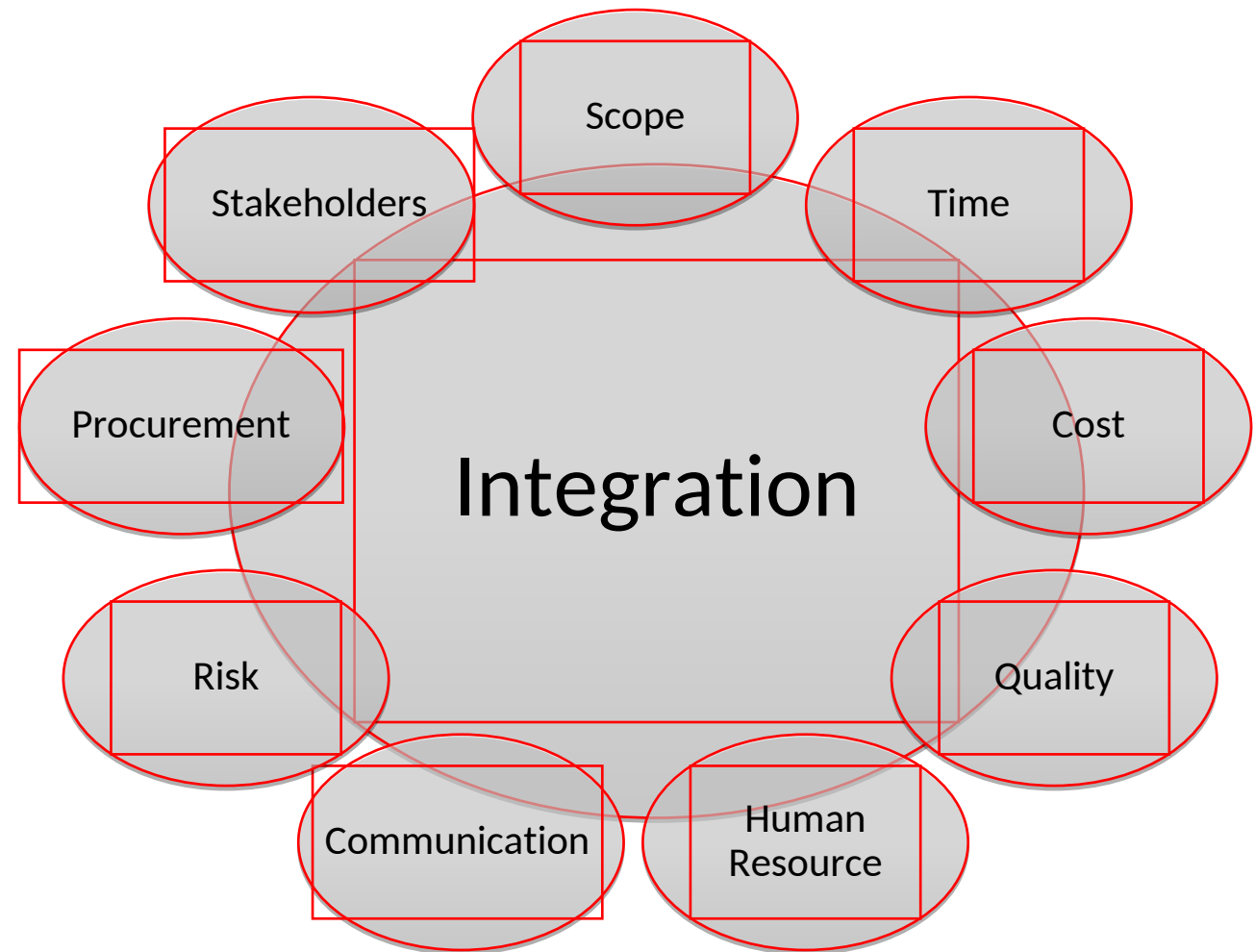
- Manage relationships with sellers,
- monitor contract performance,
- make changes if needed,
- close the contracts
- Main outputs: Work performance info, change request, procurement documentation updates & closed procurement.



Knowledge of a PM

1.4 Knowledge & Attributes of a Project Manager

10 Knowledge Areas



Attributes of a PM

1.4 Knowledge & Attributes of a Project Manager

What are skills needed to be a PM?

Personal characteristics

- Task oriented vs People oriented
- Charismatic

Behavioral skills

- Interpersonal skill
- Communication skills

-- **aka people skills:** tactics a person uses when interacting with others

-- PM need **to interact with project team members & stakeholders** (see “communication plan”)

-- comm skills including listening skills & ability to decode non-verbal cues

General business skills

MBA vs MSc?

Estimate project costs

Analyse Costs & Benefits

Calculate NP, PB, ROI, NPV

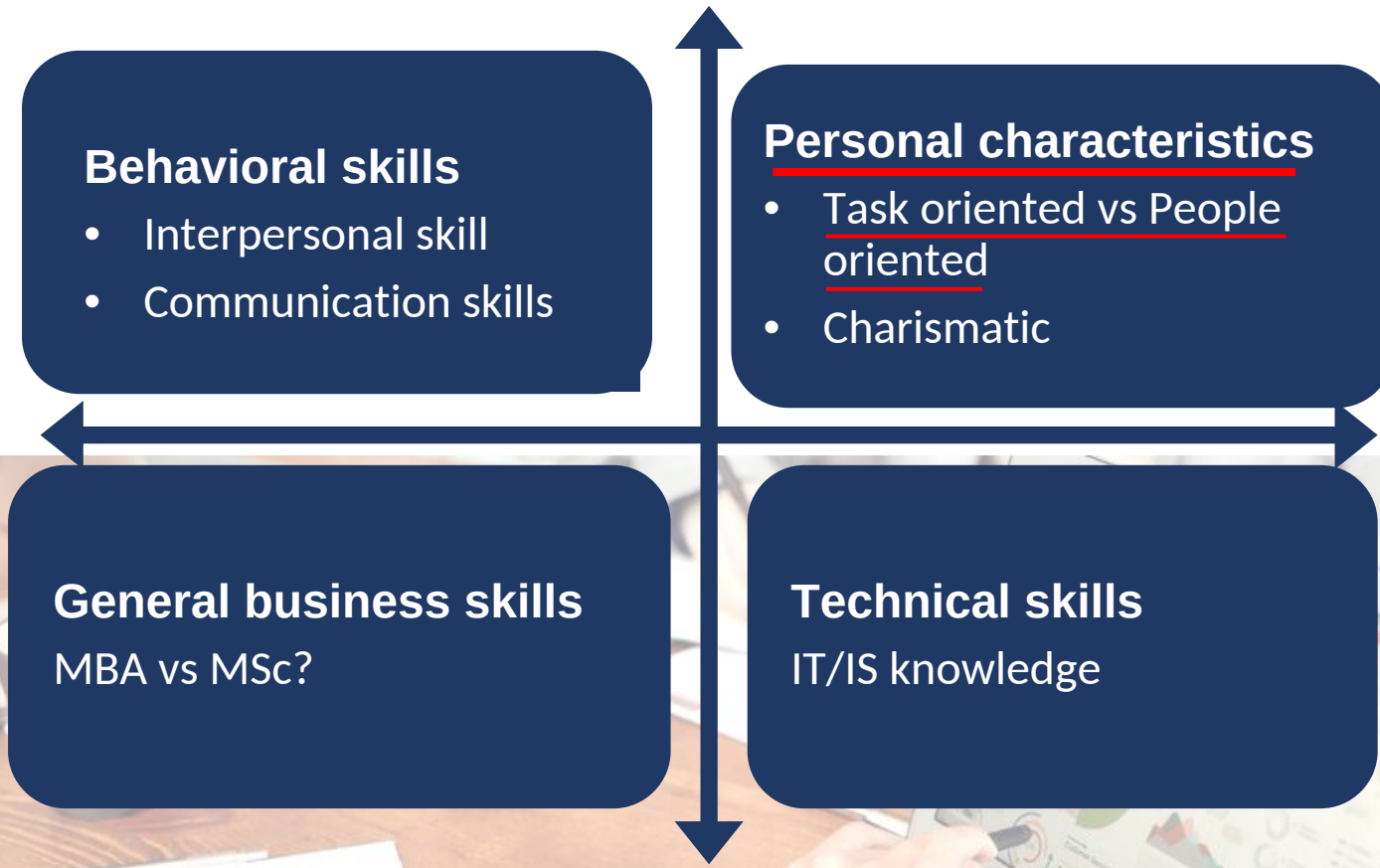
Technical skills

IT/IS knowledge



Attributes of a PM

1.4 Knowledge & Attributes of a Project Manager



Aspect	Task-Oriented	People-Oriented
Focus	Tasks & Deadlines	Team & Relationships
Communication Style	Directive (e.g. Follow these steps to complete the task)	Collaborative (e.g. what ideas do you all have to do this task?)

A project manager should be **task-oriented** to keep the project on track and be **people-oriented** to keep the team engaged and motivated.

Leading vs Managing

1.4 Knowledge & Attributes of a Project Manager

- The overriding function of management is to provide **order and consistency to organizations**, whereas the primary function of leadership is to produce **change and movement**.
- Leadership can be defined as the process of moving a group(s) in some direction through mostly non-coercive means
- Effective leadership is defined as leadership that produces movement in the long-term best interests of the group(s) (Kotter, 1988).
- Thus, leadership is defined as the ability to influence people toward the attainment of goals



1.5 Project Management Failure

Project Objectives Vs Business Objectives
Causes of Project Failures
Critical Success Factor (CSF)



Introduction

1.5 Project Management Failure

Project Objectives

Is the **targets** that the **project team** is expected to achieve such as delivering:

- the agreed functionality,
- to the required level of quality,
- on time and within budget

Business Objectives

For all the resources & efforts put in, **Return generated > costs**

3 Scenarios

1. A project **meet** the project objectives and **meet** the business objectives
2. A project **meet** the project objectives but **fail to meet** the business objectives
3. A project **fail to meet** the project objectives but successfully **meet** the business objectives

Introduction

1.5 Project Management Failure

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E.g. a computer game is delivered on time and within budget but then **not selling well**.

E.g. a computer game is completed few months late and over budget but somehow **benefits gained > the costs**

Conclusion: Which projects you will considered as successful?

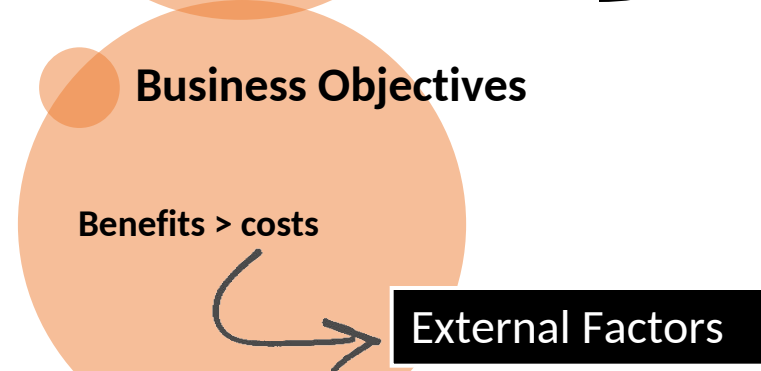
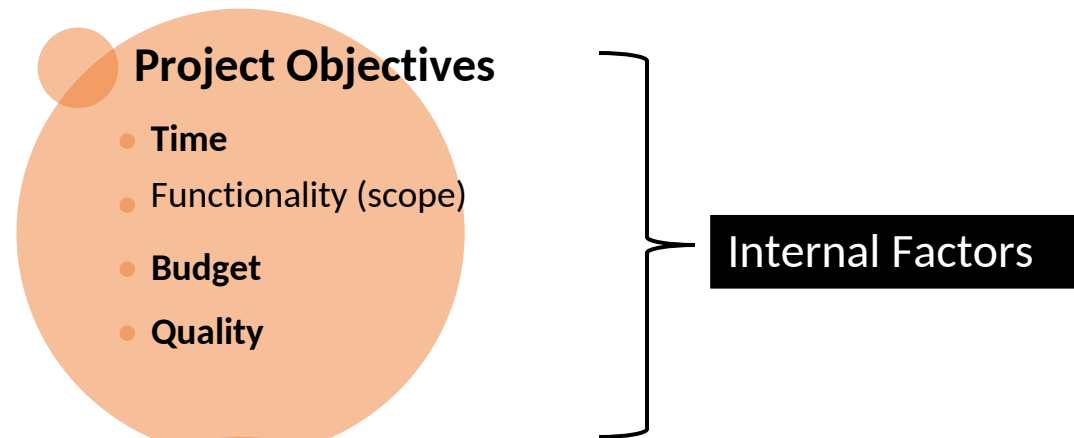
Project Failure vs Business Failure

1.5 Project Management Failure

- **A project is considered fail when:**
 - It does not satisfy the **stakeholders requirements**
 - It does not meet the **project objectives** – time, budget, quality
- A project can be successfully delivered but then be a **business failure**
- On the other hand, **a project could be late and over budget (fail to meet project objectives) but** it generates benefits that outweigh the costs (**successfully meet the business objectives**).
- **A computer game is delivered on time and within budget but then not selling well is a kind of biz risk.**
 - How to reduce business risks?**
 1. Carry out market survey and generate
 2. Carry out competitor analysis
 3. Develop prototype and let potential users test drive the prototype

Project Failure vs Business Failure

1.5 Project Management Failure



(E.g. a computer game is delivered on time and within budget but then **not selling well**)

How to reduce business risks?

- Market Survey
- Competitor Analysis
- evaluation by potential users.
- etc.

Causes of Project Failure

1.5 Project Management Failure

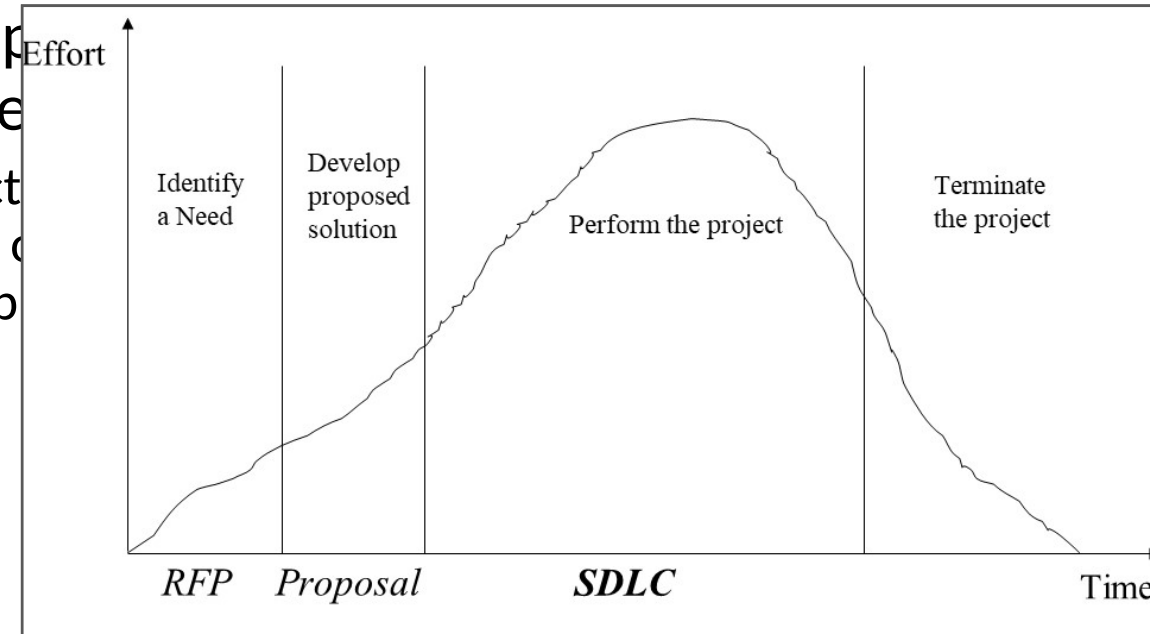
- Inadequacies or defects in project management will contribute to project failures.
- Failures in project management may be categorized according to 3 levels:
 - a. Level 1 - Failures in the Project Management Context
 - b. Level 2 - Failures in the Project Management System
 - c. Level 3 - Failures in Planning and Control Processes



Level 1: Failures in the Project Management Context

1.5 Project Management Failure

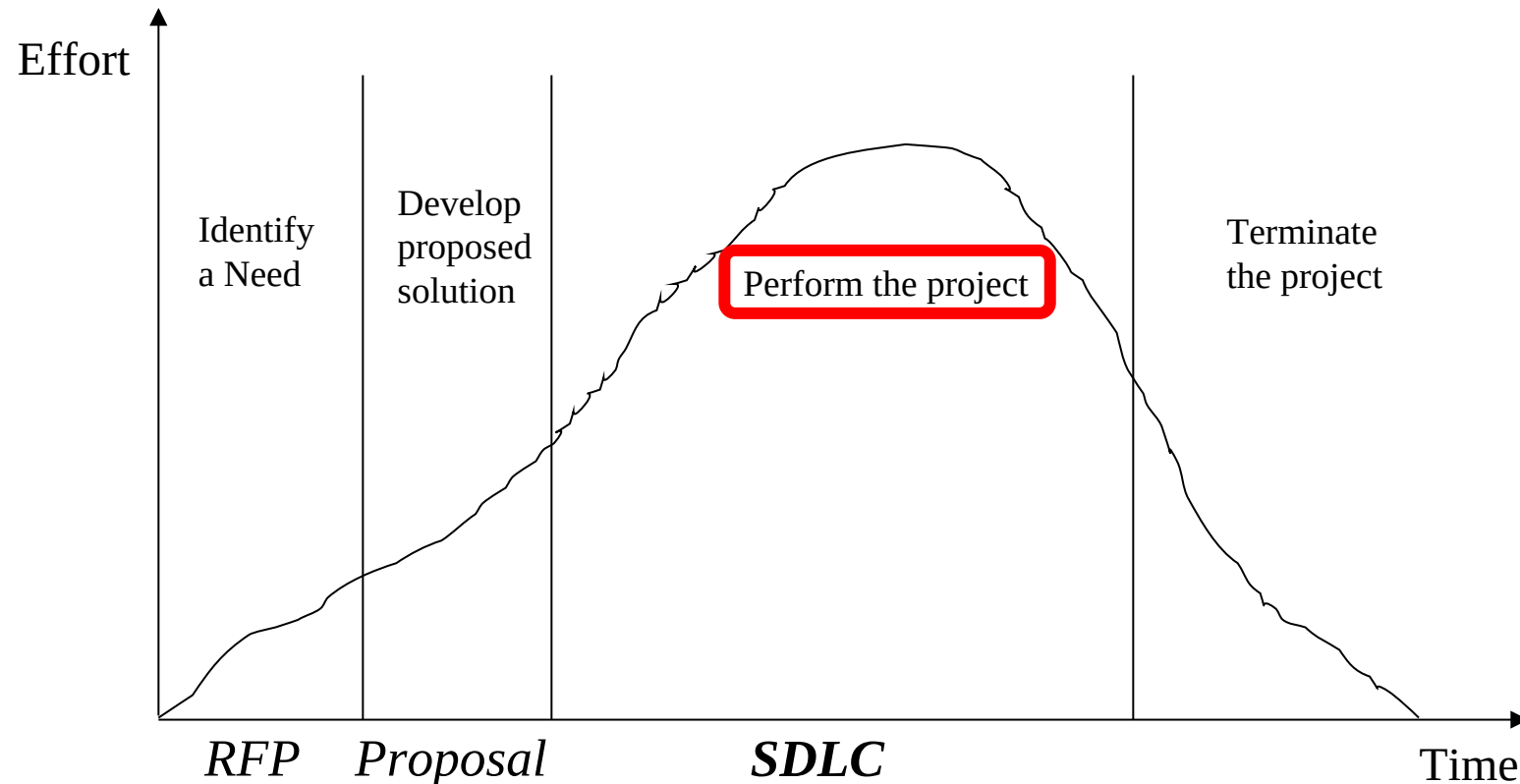
- Inappropriate “fit” of project organisation to project management, and environment at large. For example
 - A game software needs project organizational structure to promote it after accomplished. However, members of the organization are assigned without regards to proper skills and experience. Top management gives responsibility or authority to the project manager.
- **Not using appropriate project process model**
 - E.g. **Prototyping** is used in a large and complex system development which cannot capture the user requirements effectively.
- **Lack of top management support**
 - E.g. Lack of support from the financial department (**difficult to get funding from the organization**).
 - Lack of support from the sales and marketing department (**no personnels to promote the new games software**)



Level 2: Failures in the Project Management System

1.5 Project Management Failure

- Traceable to project leadership, philosophy and practice
- **Include wrong project manager**, neglect of systems approach in the project life cycle, and misuse of project management techniques
- **Examples:**
 - the **project manager is unable to control conflicts**
 - Hardware, software, resources and facilities are viewed independently without relation to their overall project



Level 3: Failures in Planning & Control Processes

1.5 Project Management Failure

- Include **inadequate communication and planning, poor estimation, poor scheduling and lack of user involvement**
(lack of user involvement will results in ... not meeting users needs)
- Examples:
 - o Information about scope, objectives, responsibility and acceptance criteria not documented early in the project
 - o The behaviour of the project team discourages user involvement

Other factors that contribute to project failure

1.5 Project Management Failure

- PM too optimistic
- Poor team development
- Under-allocation of resources

*(situation where there are **insufficient resources (people, equipment or budget)** allocated to a project activity) – will impact on project schedule, quality, ___ .*

- to achieve its objectives within the specified timeframe and quality
- Organization culture e.g. **no team work, no sharing of information between employees**
- External factors e.g. government enforce new laws

1.6 Critical Success Factors (CSFs)

1.5 Project Management Failure

- A key area where satisfactory performance is required for the organization to achieve its goals
- A means of identifying the tasks and requirements needed for success
- At the lowest level, CSFs become concrete requirements
- A means to prioritize requirements

- **Make sure you have the support of top management.**
- **Manage the project scope effectively.**
- **Competent and experienced project teams**
- ---
- **Etc**

Summary

1.1 Introduction to **Project Management**

1.2 **Project Life Cycle**

1.3 **Project Appraisal**

1.4 **Knowledge and Attributes** of a Project Manager

1.5 Project Management Failure

Project Objectives Vs Business Objectives

Causes of project failures

1.6 Critical Success Factors (CSFs)

...

...

End of chapter 1